Economic Fibrillation: Business-Cycle Synchronization and the Loss of Diversification in Global Recessions

Siddhant Shinde

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Abstract

When the business cycles of many economies line up, a shock common to all of them stops cancelling in world output. We measure that alignment for 31 economies over 1970–2022, treating each economy's 2–9 year cycle in real GDP per capita as a phase oscillator, and relate it to the five global recessions. Synchronization and swing size move together (detrended correlation +0.646). In recessions co-movement supplies 84.3% of the extra variance of world output, with correlation and larger swings contributing about equally: the 31 economies behave like about 3.5 independent ones on average and like one in 2020. Their correlation network has tightened 1.6× since the 1980s. With phases from past data only, synchronization is higher in the year before recessions ($p = 0.038$), but this rests on two events, 2009 and 2020. The variance of synchronization rises before recessions without its autocorrelation rising, so there is no critical slowing down; as an interpretation, this fits noise-induced rather than bifurcation-induced tipping. Simulated worlds with known coupling show which tests can be trusted. We make no forecast.

Keywords: business-cycle synchronization; global recessions; Kuramoto order parameter; diversification; tipping points

1 Introduction

Global recessions arrive everywhere at once. In 2009, 25 of the 31 economies studied here contracted, and in 2020, 28; in the recessions of 1982 and 1991 the number was 9 and 9. A standard argument says this should be rare. If disturbances to economies are independent, they cancel in the aggregate, and world output should vary far less than any one economy; the argument is usually traced to Lucas (1977), and it implies that aggregate volatility shrinks like one over the square root of the number of units. Two well-known results show when it fails. Gabaix (2011) shows that it fails when units are unequal in size: if a few firms are very large, their own shocks do not average out. Acemoglu et al. (2012) show that it fails when units are unequal in network position: if a few sectors supply many others, their shocks spread instead of cancelling.

This paper studies a third way the argument fails: when the units are aligned in time. National business cycles are oscillations. A disturbance common to all economies pushes an economy near the top of its cycle and one near the bottom in opposite directions, so when cycles are spread out a common shock largely cancels in world output, and when they are aligned it adds up. The world's diversification then depends not only on how many economies there are, or how large or central they are, but on how synchronized their cycles happen to be. We ask whether that synchronization rises in and before global recessions, and whether recessions look like the approach to a tipping point.

We treat each economy's 2–9 year cycle in real GDP per capita as a phase oscillator and measure how aligned the 31 economies are each year with the Kuramoto order parameter $r(t)$ (Kuramoto 1984; Pikovsky, Rosenblum and Kurths 2001). The economies produce about 81% of world output over 1970–2022. Global recessions are the five dated from world output by Kose and co-authors and the World Bank: 1975, 1982, 1991, 2009 and 2020 (Kose and Terrones 2015; Kose, Sugawara

and Terrones 2020; World Bank 2020). The question differs from that of the world business-cycle factor (Kose, Otrok and Whiteman 2003), which measures how much economies move together on average; we ask when they do. To ask whether recessions are tipping points we use the taxonomy of Ashwin et al. (2012): bifurcation-induced tipping announces itself through critical slowing down (Scheffer et al. 2009), while noise-induced tipping does not.

We find five things. (1) Synchronization and the size of the swings move together (detrended correlation +0.646). (2) In recessions the world economy loses diversification: co-movement supplies 84.3% of the extra variance of world output, with correlation and swing size contributing about equally — amplification transmitted by alignment. In the language of the effective number of independent economies, the 31 behave like about 3.5 on average and like about 1.1 in 2020. (3) The correlation network of the business cycles has tightened about 1.6× since the 1980s. (4) Synchronization rises in the year before global recessions, even with phases built from past data only (+0.379, $p = 0.038$), but this rests on two events, 2009 and 2020. (5) There is no critical slowing down: the variance of synchronization rises before recessions while its autocorrelation does not, which, as an interpretation, is what noise-induced rather than bifurcation-induced tipping would show.

We are as explicit about what we do not find. No lag of the lead–lag profile survives a correction for multiple testing; the fall in synchronization after recessions is fragile and not claimed; the share of the recession excess due to alignment changes with the sample. Simulated worlds with known coupling show that dating recessions from world output builds some synchronization into them, which is why the lead claim rests on phases from past data only, and that a synchrony-driven escape mechanism can leave a lead of this kind but rarely one as large as the real one (§5.4). The paper makes no forecast. Every test, its direction and its sample were written down and committed before it was run, and every result, null or not, is reported.

Section 2 describes the data and §3 the methods, including the size of every test on simulated worlds. Section 4 gives the five results, §5 discusses what they mean, with the arithmetic of loss of diversification in Box 1, and §6–7 give the limitations and the work that remains.

2 Data

We use annual real GDP per capita from the Maddison Project Database (2023 release; Bolt and van Zanden 2024), PPP-adjusted, for a fixed panel of $N = 31$ economies: Australia, Austria, Belgium, Brazil, Canada, China, Denmark, Finland, France, Germany, Greece, India, Indonesia, Iran, Italy, Japan, Mexico, the Netherlands, Norway, Portugal, Russia, Saudi Arabia, South Africa, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, the United Kingdom and the United States. The panel is balanced: every economy has an uninterrupted annual observation from 1950 to 2022, so no interpolation, gap-filling or unbalanced-panel weighting enters the pipeline at any stage. An earlier version of this analysis used the first 26 of these economies. China, Russia and Brazil are the largest economies missing from them, and Saudi Arabia and Iran add the oil-exporting Middle East; Russia, the second largest, was first left out by oversight, not by the inclusion rule, and was added before any test was run on it. The inclusion rule, the phase convention, the tests and their directions were written down before any test was run on the added economies (§3.7); the rule admits an economy only with a complete Maddison series for 1950–2022, which excluded the United Arab Emirates (its series misses 1991–1992). Russia's values before 1991 are back-cast estimates for the Soviet period, not market national accounts, so every test that allows it is also run with Russia's phases computed from 1992 only (§3.6.8). In Maddison's PPP figures the original 26 produced 67% of world GDP in 1970 and 51% in 2022; the 31 produce 81% and 80% (without Russia, 76% and 77%). Every result is also given without Russia, for the 30 (§3.6.8 and Appendix Table A1).

We do not assert a price base year, and none is needed. The pipeline operates on demeaned logs: any multiplicative rebasing constant κ common to a series enters as $\ln(\kappa\,\mathrm{GDPpc}) = \ln\kappa + \ln\mathrm{GDPpc}$ and is removed exactly by the subsequent mean subtraction. Every quantity in this paper — cycles, phases, correlations, order parameter — is therefore invariant to the choice of base year and to PPP

rebasing.

Two distinct date ranges are used deliberately and should not be conflated. Filtering and phase extraction are performed on the full 1950–2022 series, so that the band-pass filter and the Hilbert transform have run-in on both sides of the analysis window. All statistics are then computed on the **1970–2022** window ($T = 53$ annual observations), discarding the first two decades. This protects every reported quantity from the filter and Hilbert edge transients, which are severe in short annual records. It is a consequential choice rather than a formality: recomputing phases on a 1970–2022 series alone, with no run-in, changes the order parameter materially in individual years, shifts the synchronization–amplitude coefficient of §3.5.2 by roughly 0.1 and moves the lead–lag profile of §3.5.3 (§3.6.2).

3 Methods

3.1 Cycle extraction

For each economy i we take the natural log of real GDP per capita, $y_i(t) = \ln \mathrm{GDPpc}_i(t)$, and remove its sample mean. We isolate business-cycle frequencies with a second-order Butterworth band-pass filter applied by forward–backward filtering (`filtfilt`), which is zero-phase and therefore introduces no artificial lead or lag between economies — a necessary property, since the phase relationships between economies are the object of study.

The pass-band is 2–9 years. In normalized frequency units at an annual sampling rate ($f_{\mathrm{Nyq}} = 0.5$ yr^{-1}) the band edges are $f_{\mathrm{low}} = 1/9$ yr^{-1} and $f_{\mathrm{high}} = 0.49$ yr^{-1}. The upper edge is set at 0.49 rather than the nominal 0.5 because a Butterworth design requires the critical frequency to lie strictly inside the open interval $(0, f_{\mathrm{Nyq}})$; the resulting cycle band is thus effectively 2.04–9 years. The band brackets the conventional business-cycle range and is not tuned to any outcome; §3.6.1 reports the sensitivity of every result to the band edges.

Write the filtered cycle for economy i as $c_i(t)$. We form the analytic signal by Hilbert transform,

$$z_i(t) = c_i(t) + \mathrm{i}\,\mathcal{H}[c_i](t) = A_i(t)\, e^{\mathrm{i}\theta_i(t)},$$

which yields an instantaneous phase $\theta_i(t) = \arg z_i(t)$ and an instantaneous amplitude $A_i(t) = |z_i(t)|$ for each economy and year.

3.2 Synchronization measure

Treating the 31 economies as a population of phase oscillators, we measure collective synchronization by the Kuramoto order parameter

$$r(t)\, e^{\mathrm{i}\psi(t)} = \frac{1}{N} \sum_{i=1}^{N} e^{\mathrm{i}\theta_i(t)},$$

where $r(t) \in [0, 1]$ is the modulus of the mean phase vector and $\psi(t)$ is the collective phase. $r = 0$ corresponds to uniformly dispersed phases and $r = 1$ to identical phases. We also define the mean cycle amplitude $\bar{A}(t) = N^{-1} \sum_i A_i(t)$, a measure of how large the swings are, independent of whether they are aligned. Throughout, $A_i(t)$ denotes a single economy's instantaneous amplitude and $\bar{A}(t)$ the cross-sectional mean; no other amplitude symbol is used.

Because $r(t)$ is a modulus, it is non-negative and biased away from zero in finite samples; with $N = 31$ its root-mean-square under independent uniform phases is exactly $N^{-1/2} \approx 0.18$ (its mean is about 0.16). We therefore never interpret the level of r against zero, only its variation over time and against matched surrogates (§3.4).

3.3 Crisis dating

Crisis years are taken exogenously from the Kose–Terrones definition of a **global recession** (Kose and Terrones 2015; Kose, Sugawara and Terrones 2020), which identifies contractions in world real GDP per capita accompanied by broad-based declines in other measures of global activity. This yields five dates in our window: **1975, 1982, 1991, 2009 and 2020**. The first four are those of Kose, Sugawara and Terrones (2020), written before the pandemic; 2020 is dated as a global recession under the same definition by the World Bank (2020). These dates are fixed before any statistic is computed and are never adjusted, re-weighted, or selected on the basis of the synchronization series. We use only the crisis *onset* years; no crisis is assigned a duration.

The count matters for interpretation and is stated plainly: five events in 53 years. Every crisis-conditioned test in this paper therefore rests on five degrees of freedom at most, and the leave-one-out analysis of §3.6.3 shows that the effective count for the pre-crisis result is smaller still.

3.4 Null models

Two nulls are used, for two different purposes, and a third is used where the end of the sample matters (§3.4.4).

3.4.1 Circular-shift permutation. For any test that correlates a continuous series with the crisis indicator, we build the null by circularly shifting the crisis indicator vector G — a binary vector with ones at the five crisis years — through all $T - 1 = 52$ non-identity rotations, recomputing the statistic each time (53 configurations in total, counting the unshifted one). This preserves both the autocorrelation of the continuous series and the number and relative spacing of the crisis dates, and destroys only their alignment with the series. The p-value is the fraction of shifts whose statistic equals or exceeds the observed one, with the usual $+1$ correction in numerator and denominator.

This null has a hard resolution limit that must be quoted alongside any p it produces: with 52 shifts, the smallest attainable p-value is $1/53 = 0.019$. A reported $p = 0.019$ therefore means "no shift beat the observed value" and cannot be read as evidence of a smaller effect size than $p = 0.019$; it is a floor, not a measurement.

3.4.2 Spectrum-preserving surrogate economy sets. For tests concerning the structure of the oscillator population, we generate 5,000 surrogate 31-economy sets by Fourier phase randomization (Theiler et al. 1992). We take the real FFT of each economy's band-passed cycle, retain the amplitude spectrum exactly, replace each non-trivial Fourier phase with an independent draw from $U(0, 2\pi)$ (holding the DC and, for even record length, Nyquist phases real), and invert the transform. Each surrogate economy therefore has the same power spectrum, and hence the same autocorrelation and cycle periodicity, as its real counterpart, but its phase relationship to every other economy is destroyed. Phases are randomized independently across economies, so the surrogate ensemble is the null of *spectrally identical but mutually unsynchronized* economies. Surrogate p-values with 5,000 draws have a floor of $1/5001 = 2 \times 10^{-4}$. Surrogates are drawn from each economy's 1950–2022 cycle and pass through the same Hilbert transform and 1970–2022 window as the data, so real and surrogate phases share one convention (§3.6.2).

3.4.3 Size of the tests on simulated worlds. To check that each test rejects at its nominal rate when there is nothing to find, we ran every test on 1,000 simulated 31-economy worlds of each of two kinds — phase oscillators (Kuramoto) and phase-and-amplitude oscillators (Stuart–Landau) — with no coupling between economies and no common forcing. Each economy keeps its observed mean period, swing size and trend, and the noise is set so that the year-to-year spread of phase steps matches the data; the simulated series pass through the same band-pass, Hilbert transform and tests as the data, with 199 surrogates per test. Table 1 gives the share of worlds in which each test rejects at $p \leq 0.05$. With the real recession dates, which are unrelated to a simulated world, a correct test rejects in 3.8% of worlds when its null is the circular shift (below 0.05 only $1/53$ and $2/53$ are possible) and in 5% when it is 199 surrogates. Of the 20 tests, 17 stay at or below that rate in the Kuramoto worlds and 17 in the Stuart–Landau worlds. The exceptions are the swing-size term of the identity (13.8%–18.9%), which leans on the last years of the sample and is checked against

surrogates with ends in §3.4.4; the circular-shift test of r against swing size (5.7%–6.2%); and, in one kind of world each, the lag -2 arc (6.2%) and the alignment term (6.5%). At least one of the five arc lags rejects in 17.3%–18.4% of uncoupled worlds, the multiplicity of §3.6.4 in numbers. With recession years taken instead from each simulated world's own output, the way global recessions are dated, the two-sided crisis-year and lead-1 arcs reject in 8.2%–8.9% and 8.5%–8.6% of uncoupled worlds (Limitations).

Table 1. Share of 1,000 uncoupled simulated worlds in which each test rejects at $p \leq 0.05$. **Bold**: the 95% interval lies above the correct rate. Real dates: 1975, 1982, 1991, 2009, 2020. Own dates: each world's five years of lowest growth in world GDP per capita, at least five years apart.

Test	Correct rate	Kuramoto, real dates	Stuart–Landau, real dates	Kuramoto, own dates	Stuart–Landau, own dates
Arc, lead-2	3.8%	3.3%	3.6%	**5.2%**	**5.3%**
Arc, lead-1	3.8%	2.9%	4.4%	**8.5%**	**8.6%**
Arc, crisis year	3.8%	4.4%	3.3%	**8.9%**	**8.2%**
Arc, after-1	3.8%	4.5%	3.6%	4.2%	2.6%
Arc, after-2	3.8%	**6.2%**	4.6%	3.9%	4.3%
Arc, lead-2, past-only phases	3.8%	4.1%	3.4%	3.3%	3.5%
Arc, lead-1, past-only phases	3.8%	3.5%	3.8%	2.9%	2.3%
r with swing size (shifts)	3.8%	**6.2%**	**5.7%**	**6.2%**	**5.7%**
r with swing size (surrogates)	5.0%	5.5%	5.4%	5.5%	5.4%
Cross term at recessions, GDP weights	5.0%	5.3%	5.4%	**8.9%**	**10.8%**
Cross term at recessions, equal weights	5.0%	4.6%	4.6%	**6.8%**	**6.8%**
Identity: alignment at recessions	5.0%	4.8%	**6.5%**	**11.4%**	**12.1%**
Identity: swing size at recessions	5.0%	**13.8%**	**18.9%**	**7.5%**	**14.9%**
MST contraction before recessions	5.0%	2.8%	4.1%	5.0%	3.8%
MST contraction, detrended	5.0%	3.0%	4.5%	5.1%	3.7%
MST fragility before recessions	5.0%	4.9%	5.2%	5.3%	5.1%
Phase leads clustered	5.0%	3.0%	5.3%	3.0%	5.3%
Slowing down: autocorrelation of r	5.0%	5.1%	4.2%	5.7%	4.5%
Slowing down: variance of r	5.0%	5.1%	6.1%	5.0%	5.5%
Slower recovery after recessions	5.0%	4.8%	4.5%	5.2%	4.0%
Any of the five arc lags	—	18.4%	17.3%	26.2%	25.0%
Any of lead-2, lead-1, crisis	—	9.9%	10.2%	20.2%	20.1%

3.4.4 Surrogates with ends. The surrogates of §3.4.2 are circular: they have no first or last year, so the band-pass and the Hilbert transform treat their ends differently from the data's, where the last years of the sample are the least reliable. For every test whose statistic uses a year from 2019 to 2022 — the crisis year 2020, a lead or lag reaching those years, or a window ending in them — we also compare the statistic with surrogates that have ends. Each economy's growth rates are given random Fourier phases on a series eight times longer than the sample, with the same periodogram; a 72-year stretch from a random start is cumulated from the real 1950 level and passed through the same band-pass, Hilbert transform and windows as the data. For tests whose null is independent surrogates the phases are drawn independently for every economy; for the lead–lag arcs, whose null is the circular shift, one random phase per frequency is shared by every economy, which keeps their co-movement and so the level and persistence of $r(t)$ and randomizes only the timing. We draw 5,000 such surrogates (floor 0.0002). On uncoupled simulated worlds (§3.4.3) these nulls reject at about 5%, including the swing-size term (1.5% and 4.8%). On the 31 economies no result we claim changes side of 0.05: the swing-size term and r against swing size stay at the floor ($p = 0.0002$), the variance rise of §3.5.7 is $p = 0.029$, and the two-sided lead-1 moves from the circular-shift floor 0.019 to $p = 0.033$ (0.043 on the 30, 0.042 on the 26). In all, 6 p-values of sensitivity or unclaimed results move across 0.05 (Appendix Table A4); the one that matters for the text is the lag -2 arc on the 31 (0.057 → 0.039; Limitations).

3.5 Tests

3.5.1 Loss-of-diversification decomposition. The world output cycle is the weighted aggregate of the 31 band-passed economy cycles,

$$C_w(t) = \sum_{i=1}^{N} w_i \, c_i(t),$$

with weights $w_i \geq 0$, $\sum_i w_i = 1$. On each rolling 10-year window its variance decomposes exactly into an own-term and a cross-term,

$$\mathrm{Var}[C_w] \;=\; \underbrace{\sum_{i} w_i^2 \, \sigma_i^2}_{\text{own}} \;+\; \underbrace{\sum_{i \neq j} w_i w_j \, \sigma_{ij}}_{\text{cross}},$$

where σ_i^2 is economy i's within-window cycle variance and σ_{ij} the pairwise covariance. Variances are computed with the population convention (dividing by the window length $W = 10$). The own-term is what world output would fluctuate by if the 31 economies were uncorrelated; the cross-term is the co-movement surcharge. The decomposition is an identity, not a model, and holds window by window.

The test is deliberately *not* "does world variance rise with average correlation." That is close to algebraically guaranteed: for equal weights and equal cycle variances the cross share is cross/total $= (N-1)\bar{\rho}/(1+(N-1)\bar{\rho})$, so with $N = 31$ the 465 covariance pairs outnumber the 31 variances by more than an order of magnitude and the cross-term dominates at any $\bar{\rho}$ above a few hundredths. Observed $\bar{\rho}$ ranges from 0.08 to 0.77 across our windows, which alone would force a cross share between 0.69 and 0.96 if the economies swung equally. Unequal swings lower the share wherever one economy dominates the own-term: the equal-weighted share ranges from 0.19 to 0.95 and stays below 0.40 in 9 of the 12 windows ending 1985–1996, when Iran's cycle alone supplies between 19% and 42% of the own-term. Reporting a high cross share as a finding would therefore still be reporting arithmetic. The non-circular question is which term supplies the *excess* variance in global-recession years relative to normal years:

$$\Delta_{\text{own}} = \langle \text{own} \rangle_{\text{crisis}} - \langle \text{own} \rangle_{\text{normal}}, \qquad \Delta_{\text{cross}} = \langle \text{cross} \rangle_{\text{crisis}} - \langle \text{cross} \rangle_{\text{normal}},$$

summarized as the cross-term's share of the excess, $\Delta_{\text{cross}}/(\Delta_{\text{own}} + \Delta_{\text{cross}})$.

Two weightings are computed and both are reported. **GDP-weighted** (w_i proportional to the economy's within-window mean real GDP, i.e. GDP per capita times population) is the primary framing, since it is the actual world economy. **Equal-weighted** ($w_i = 1/N$) is the robustness check and ties directly to the equal-weight order parameter of §3.2, which gives every economy one vote. Windows may start in 1962 and never earlier (§3.6.7), so the decomposition is defined from 1971 and all five global recessions enter it. The 1975 window (1966–1975) draws on run-in years; the version with windows kept inside 1970–2022, which loses 1975, is reported in §3.6.7.

Across 1971–2022 the cross-term supplies the great majority of the excess crisis variance under both weightings: 84.3% GDP-weighted and 97.0% equal-weighted (S30: 83.3% and 96.8%). The underlying cross-shares are 0.706 in crisis windows against 0.620 in normal windows (GDP-weighted), and 0.783 against 0.691 (equal-weighted). The direction is consistent across weightings, and the gap is a shift of about 9% on a base of about two-thirds. The equal-weighted figure is the larger because equal weights give the 465 covariance pairs their maximum leverage; the GDP-weighted figure, in which a few large economies carry most of the own-term, is the one we treat as describing the world economy.

The cross-term is not the same thing as correlation: it rises when economies move together and when they swing harder. Changing only the pairwise correlations from their normal-window means to their crisis-window means (mean correlation 0.291 → 0.385) supplies 46% of the cross-term's crisis rise GDP-weighted (50% equal-weighted); changing only the cycle sizes (mean cycle s.d. × 1.10) supplies 40% (37%); their interaction supplies the remaining 14% (13%). On the 30

without Russia the split is 44%, 41% and 15%. The two parts are of similar size, and which is larger depends on the sample: on the original 26 economies the larger swings supplied more (53% against 33%; §3.6.8). Co-movement and larger swings contribute about equally.

Significance is assessed against the spectrum-preserving surrogate ensemble of §3.4.2. §3.6.6 reports the outcome of that test, which does not behave as a well-posed significance test for the share statistic, and states what we do and do not claim as a result.

3.5.2 Synchronization–amplitude coupling. We correlate $r(t)$ with $\bar{A}(t)$ over 1970–2022 by Pearson correlation, in two versions: on the raw series, and after removing a slow trend from each by subtracting a Gaussian-smoothed copy ($\sigma = 8$ yr). The detrended version is the headline: both series carry secular drift over five decades, and the raw correlation cannot distinguish co-movement at business-cycle timescales from two coincident long-run trends. The two versions agree in sign and significance — +0.472 raw against +0.646 detrended, both at the floor $p = 0.019$ (S30: +0.458 and +0.641) — so the result does not hinge on the detrending choice; we report the detrended value because it isolates business-cycle timescales, not because it is the larger. Significance is assessed by the circular-shift null of §3.4.1 applied to the paired series, and separately against the surrogate ensemble of §3.4.2 ($p = 0.0002$, the ensemble floor). This test uses no crisis dates at all.

3.5.3 Crisis-relative lead–lag profile. For each lag $\ell \in \{+2, +1, 0, -1, -2\}$ we correlate $r(t)$ with the crisis indicator shifted by ℓ, so that positive ℓ tests whether synchronization *precedes* crises and negative ℓ whether it falls after them. Leading lags are tested one-sided upward and trailing lags one-sided downward, matching the pre-registered direction of the hypothesis; the contemporaneous lag is tested upward. Significance uses §3.4.1. Because the band-pass and the Hilbert transform both use data after year t, a lead found this way is not a forecast; every lead is re-tested with phases built from past data only (§3.6.2).

Five lags are tested, and the raw p-values are reported as such alongside a multiplicity correction over the five (§3.6.4). We report the raw values because they are the natural output of the permutation scheme, and the corrected values because five lags were in fact examined.

3.5.4 Composite crisis epoch. We average r and $\bar{A}$ over a ± 4-year window centred on each of the five crisis onsets and take the mean across events, giving a composite epoch profile for each quantity. The peak lag of each profile is read off directly. This is a descriptive alignment, not a test, and is reported with no p-value.

3.5.5 Network geometry. On 10-year rolling windows we compute the 31×31 Pearson correlation matrix of the band-passed cycles, convert it to the standard ultrametric-compatible distance $d_{ij} = \sqrt{2(1 - \rho_{ij})}$ (Mantegna 1999), build the minimum spanning tree by Prim's algorithm, and record its mean edge length $L(t) = (N-1)^{-1} \sum_{e \in \mathrm{MST}} d_e$. Falling L indicates a tightening network. Windows may start in 1962 and never earlier, so that each has at least twelve years of filter run-in (§3.6.7); $L(t)$ is therefore defined from 1971. Two questions are asked of L: whether it trends secularly — reported as a description via decade means, 0.687 in 1979–89 (the first decade whose windows all lie inside the 1970–2022 analysis window) against 0.424 in 2013–2022, a factor of 1.6× (S30: 0.680 against 0.421, also 1.6×) — and whether contraction coincides with the pre-crisis windows. The latter is quantified by a Kendall-τ trend over the 8 years preceding each onset, averaged over crises (1975's window has four of its eight years defined), and tested one-sided downward against the surrogate ensemble of §3.4.2, both on raw L and on L minus a Gaussian-smoothed copy of itself ($\sigma = 8$ years). Both are null: $\tau = -0.157$, $p = 0.206$ on raw L and $\tau = +0.014$, $p = 0.561$ detrended (S30: -0.143, $p = 0.219$; $+0.057$, $p = 0.659$). The test is described here so that it is on record as having been run and failed, not inferred from its absence.

3.5.6 Phase-lag structure. For each economy we compute its mean phase offset from the collective phase, $\langle \arg e^{\mathrm{i}(\theta_i - \psi)} \rangle$, and convert it to years by dividing by the mean angular frequency $\bar{\omega}$ (estimated as the mean year-on-year phase increment across economies, giving a mean cycle period of 5.93 years). An economy with a consistently positive offset sits ahead of the collective phase; whether that constitutes economic leadership is a question for later work and is not addressed here. The test statistic is the standard deviation of these 31 lags, observed at 0.546 years: a large spread would leave leader–follower structure recoverable, a small spread implies the economies move near-

simultaneously. It is compared to the surrogate ensemble of §3.4.2. Because the scientific question is whether the observed spread is *smaller* than chance, the test is one-sided **downward**: the observed 0.546 yr lies below all 5,000 surrogate values (surrogate mean 0.992, minimum 0.554), giving $p = 0.0002$, the ensemble resolution floor (S30: 0.535 yr, $p = 0.0004$). The economies are phase-locked tightly enough that no epicenter is recoverable.

3.5.7 Critical slowing down. We test the standard early-warning battery (Scheffer et al. 2009) — rising lag-1 autocorrelation and rising variance of the fluctuations of $r(t)$ around a Gaussian-smoothed trend ($\sigma = 5$ years), on trailing 15-year windows — over the 8 years before each onset, with the trend measure and surrogate test of §3.5.5, one-sided upward. A third indicator often added, the restoring time, was considered and dropped: estimated as $1/(-\ln\varphi)$ from the same lag-1 autocorrelation φ, it is an increasing function of φ, so its trend test repeats the autocorrelation's and adds no independent evidence. We complement the two indicators with a test that treats coupling as the bifurcation parameter: inverting the mean-field Kuramoto relation $r = \sqrt{1 - K_c/K}$ gives an implied coupling $K/K_c = 1/(1-r^2)$, and critical slowing down would show as a fall in the recovery rate $\lambda = -\ln\varphi$ in the years just after each onset relative to all other years, tested one-sided against the surrogate ensemble of §3.4.2. Windows may start in 1962 (§3.6.7), so the pre-crisis windows of 1982 (six of eight years defined), 1991, 2009 and 2020 enter; 1975 has none.

Two of the three tests are null. Lag-1 autocorrelation shows no pre-crisis rise ($\tau = -0.152$, $p = 0.789$), and the recovery rate does not fall after onsets (change +0.002, $p = 0.502$). The variance of $r(t)$ does rise before crises ($\tau = +0.438$, $p = 0.031$; S30: +0.420, $p = 0.037$), but without the rise in autocorrelation. Critical slowing down raises both together, because both follow from a slower return to equilibrium; variance alone also rises when the shocks that move $r(t)$ grow larger. A rise in variance without a rise in autocorrelation is therefore not the slowing-down signature. It is what loss of diversification (§3.5.1) would produce: when shocks to the economies stop cancelling, $r(t)$ swings more widely without returning more slowly. This is a reading of the result, not a separate test. The variance rise also rests on the two modern events, like the lead of §3.5.3: the windows before 2009 and 2020 alone give $\tau = +0.857$ ($p = 0.003$), those before 1982 and 1991 give $\tau = +0.018$ ($p = 0.513$), and dropping either 2009 or 2020 leaves it short of significance ($p = 0.206$ and 0.125). It is significant under convention P and the edge-robust E ($p = 0.045$) but not under W or N ($p = 0.135$ and 0.073; on S30, E gives $p = 0.054$; §3.6.2). Annual data over 8-year pre-crisis windows give these estimators very few effective degrees of freedom, so the nulls are weak evidence in the statistical sense and we do not read them as establishing an absence.

It is not, however, an empty result. Rising autocorrelation and variance together are the specific signature of a system approaching a *bifurcation* — the slow eigenvalue going to zero as a fixed point loses stability. In the taxonomy of Ashwin et al. (2012) this is B-tipping, and it is only one of three routes; noise-induced (N-) and rate-induced (R-) tipping produce no such precursor, because in neither case does the system's restoring rate vanish before the transition. The absence of that joint fingerprint is therefore consistent with a non-bifurcation transition, and is informative about *which class* of tipping is in play even where it is uninformative about proximity to a threshold. We report these tests as what the transition is not. The mechanism the paper proposes is loss of diversification (§3.5.1), argued in the Discussion; nothing in this paper tests whether that mechanism has a threshold.

3.6 Robustness and disclosure

Every check in this section was run on the final pipeline, and all outcomes are reported whether or not they support the paper's claims.

3.6.1 Filter band. Results are recomputed for pass-bands of 2–8 and 3–10 years. The pre-crisis lead is present in all three bands (lead-1 between +0.19 and +0.25) but its p-value moves across the conventional threshold: lead-1 gives $p = 0.019$ (2–9 yr), 0.094 (2–8 yr) and 0.019 (3–10 yr). The post-crisis decline is not robust to the band: the lag -2 coefficient is -0.26 ($p = 0.057$) at 2–9 years, -0.27 ($p = 0.019$) at 2–8 years and +0.00 ($p = 0.509$) at 3–10 years.

3.6.2 Detrending method and phase convention. Substituting an HP filter ($\lambda = 100$) for the

band-pass leaves the lead result materially unchanged (lead-1 +0.29, $p = 0.019$). Using first differences of log GDP per capita instead weakens the lead-2 term to $p = 0.057$ and eliminates the post-crisis decline entirely (+0.04, $p = 0.679$). The post-crisis result is thus method-dependent in a way the pre-crisis result is not.

The phase convention matters too. Besides the convention used throughout (P: band-pass and Hilbert transform on 1950–2022, statistics on 1970–2022) we compute four others: W, the Hilbert transform on 1970–2022 only, after the same band-pass; N, both steps on 1970–2022 only, with no run-in; E, an edge-robust version of P that mirrors each cycle by nine years (the longest period in the band) at both ends before the Hilbert transform; and C, phases from past data only, recomputing the band-pass and Hilbert transform on the data up to each year and keeping that year's phase. C uses no future data, so a lead that survives it is not an artifact of the two-sided filters.

Table 2. The arc and the synchronization–swing-size link under each phase convention (31 economies; the last column the 30). Correlation, one-sided p in brackets (circular shifts, floor 0.019).

Convention	Lead-2	Lead-1	Lag -2	r–$\bar{A}$, detrended	S30 lead-1
P	+0.235	+0.231	−0.258	+0.646	+0.217
	(0.038)	(0.019)	(0.057)	(0.019)	(0.019)
W	+0.216	+0.204	−0.165	+0.566	+0.184
	(0.057)	(0.019)	(0.113)	(0.019)	(0.057)
N	+0.216	+0.192	−0.190	+0.547	+0.173
	(0.038)	(0.057)	(0.094)	(0.019)	(0.094)
E	+0.233	+0.236	−0.160	+0.583	+0.226
	(0.038)	(0.019)	(0.132)	(0.019)	(0.019)
C	−0.044	+0.379	−0.159	+0.726	+0.369
	(0.585)	(0.038)	(0.170)	(0.019)	(0.038)

Lead-1 is significant under P, W, E and C and falls short under N ($p = 0.057$); on the 30 without Russia it also falls short under W ($p = 0.057$). Of the two leads it is the one that survives phases built from past data only, and it is the one we interpret; lead-2 does not survive them, and the post-crisis fall is significant under none of the five conventions. Under past-only phases the crisis year also reaches the floor (+0.417, $p = 0.019$, against 0.189 under P). The past-only check was pre-registered for the leads only, and we do not interpret this: a phase built from data up to the crisis year is estimated at the very end of its series, where the recession itself sits. The synchronization–amplitude link is significant at the floor under every convention, which moves its size between +0.547 and +0.726. We report P throughout, on the ground that the edge transients its run-in removes are an artifact, not because it gives the larger values.

3.6.3 Leave-one-crisis-out. Dropping each crisis in turn and refitting §3.5.3 shows that the pre-crisis lead is not evenly carried by the five events. Lead-1 survives removal of 1975, 1982 and 1991 ($p = 0.019$, 0.019 and 0.019; without 1975 it is $p = 0.057$ on the 30 without Russia), but dropping either 2009 or 2020 destroys its significance ($p = 0.226$ and 0.283), and dropping both leaves no lead at all (−0.02, $p = 0.491$). A split-sample fit tells the same story: 1970–1995 (crises 1975, 1982, 1991) shows no significant lead (lead-1 +0.24, $p = 0.154$), while 1996–2022 (crises 2009, 2020) does (+0.38, $p = 0.037$, the floor for 27 years). The effective event count behind the pre-crisis claim is therefore $n \approx 2$, both in the modern era, and the claim is stated in the paper as an era-specific property of the tightly coupled economy rather than a fifty-year regularity. The variance rise of §3.5.7 rests on the same two events.

3.6.4 Multiplicity. Across the five lags of §3.5.3, no coefficient survives a Bonferroni correction (lead-1: $p_{\text{adj}} = 0.094$; lead-2: 0.189; lag -2: 0.283) or Benjamini–Hochberg control of the false discovery rate ($q = 0.094$ for lead-2, lead-1 and lag -2). This is partly built into the test. With 53 circular shifts the smallest p the test can return is $1/53 = 0.019$, so no single lag can pass a Bonferroni correction over five lags ($5/53 = 0.094$), and a lag can pass Benjamini–Hochberg only if another lag also reaches the floor, as lead-1 and lag -2 did together on the original 26 economies ($q = 0.047$; §3.6.8). Corrected significance is therefore not what supports lead-1. Its support is that it survives phases built from past data only (§3.6.2) and the removal of any one economy (§3.6.5);

with past-only phases and China left out together it is $p = 0.057$ (Appendix Table A1), on the 31 as on the 30. It does not survive the removal of 2009 or 2020 (§3.6.3).

3.6.5 Leave-one-economy-out. Removing any one of the 31 economies leaves lead-1 between +0.22 and +0.25 with $p \leq 0.038$ (the United States, Germany and Japan: $p = 0.019$, 0.019 and 0.019). No single economy drives the result, including each of the five added in this version (§3.6.8).

3.6.6 Surrogate test of the decomposition. The share statistic of §3.5.1 does not admit a well-behaved surrogate test, and we report this rather than substituting a statistic that does. Its denominator $\Delta_{\text{own}} + \Delta_{\text{cross}}$ is a difference of window means that is near zero and of either sign under the surrogate null, so the ratio is unbounded and its null distribution is heavy-tailed and non-monotone in the quantity of interest. Against 5,000 surrogate sets, 2,390 draws equal or exceed the observed equal-weighted share and 1,581 the GDP-weighted share — nominal p-values of 0.48 and 0.32, which reflect the pathology of the ratio rather than the strength of the effect. Tested on the unnormalized components instead, Δ_{cross} exceeds every one of the 5,000 surrogates under both weightings ($p = 0.0002$, the ensemble resolution floor), and so does Δ_{own} equal-weighted (GDP-weighted it exceeds all but 191, $p = 0.038$), but that test is uninformative for our purpose: it rejects the null that crisis windows resemble phase-randomized ones at all, which is a much weaker statement than the decomposition claim, and it rejects for the own-term too. In simulated worlds with trade coupling it rejects even when the recession dates have nothing to do with the worlds (Limitations).

We therefore state the decomposition result as a descriptive magnitude with an explicit floor, not as a significance claim: the cross-term supplies the large majority of the excess crisis variance under both weightings, on five events, and the algebraic baseline of §3.5.1 already guarantees a high cross share before any economics enters. What the data add beyond arithmetic is the *gap* between crisis and normal windows (§3.5.1), which is modest in absolute terms. A defensible significance test for this statistic — one whose null distribution is proper — is not available to us and we do not report one.

3.6.7 Window convention of the rolling statistics. Every rolling statistic — the network measure (§3.5.5), the decomposition (§3.5.1) and the early-warning and coupling tests (§3.5.7) — uses windows that may start in 1962 and never earlier, leaving at least twelve years of filter run-in. As a sensitivity check each is recomputed with windows kept inside 1970–2022, dropping any pre-crisis window that is then only partly defined. For L this changes no value from 1979 onward, so the decade means of §3.5.5 are unaffected; the pre-crisis contraction test, now on 1991, 2009 and 2020, stays null ($\tau = +0.095$, $p = 0.654$ raw; +0.333, $p = 0.910$ detrended). The decomposition loses the 1975 window and gives 86.8% GDP-weighted and 96.9% equal-weighted. Lag-1 autocorrelation ($\tau = -0.107$, $p = 0.644$) and the coupling test (+0.065, $p = 0.520$) stay null; the variance trend, now on 2009 and 2020 only, is +0.857 ($p = 0.003$).

3.6.8 The original 26, the 30 and the five added economies. Every test was also run on the original 26 economies and on the 30 without Russia, under the same conventions; on the 31 with each added economy left out; with Iran and Saudi Arabia, whose cycles are the largest relative to the median of the 26, left out together; and with Russia's phases computed from 1992 only, so that its back-cast Soviet-era values cannot drive the result (Appendix Table A1). Without Russia no primary result changes sign or crosses $p = 0.05$: lead-1 is +0.217 ($p = 0.019$), the synchronization–amplitude link +0.641, the network tightens by 1.6×. With Russia's phases from 1992, lead-1 is +0.224 ($p = 0.019$) and the synchronization–amplitude link +0.655. On the 26 the lead–lag profile is almost the same (lead-1 +0.216, $p = 0.019$; with past-only phases +0.374, $p = 0.038$), lead-1 and lag -2 pass Benjamini–Hochberg together ($q = 0.047$), the network tightens by 1.8×, and the larger swings supply more of the cross-term's crisis rise than the higher correlations (53% against 33%). Leaving out one added economy moves lead-2 and past-only lead-1 to $p = 0.057$ without China; the fall after recessions (lag -2, $p = 0.057$ on the 31) crosses to $p < 0.05$ without 20 of the 31 single economies and with Russia from 1992, which marks it as sitting on the threshold rather than as stronger (§3.6.1–7.2). The alignment share of the excess in world swing is sample-sensitive: 39% on the 26, 19% on the 30, 33% on the 30 without Iran and 28% on the 31. Replacing Maddison's 1991–2011 growth for Norway, Switzerland and South Africa, which runs two to three

points above national accounts, with the Global Macro Database's changes no conclusion (on the 31, lead-2 moves to $p = 0.057$ and lag -2 to $p = 0.038$).

3.7 Reproducibility

The input data are the Maddison Project Database (2023 release) as distributed in the `MaddisonData` R package (repository `sbgraves237/maddisondata`, commit `3ed424b6`; the data file `MaddisonData.rda`, sha256 `63a3cff9…`, is unchanged since October 2025). The pipeline rebuilds from these pinned files with one command and is deterministic apart from the surrogate ensembles, which are seeded; every result is saved with its settings, seed and code version. The sample, phase convention, tests and test directions were written down and signed before any test was run on the added economies, and extended by a dated amendment before Russia was added; every pre-registered test is reported. The earlier results on the 26 economies reproduce exactly from the rebuilt pipeline under the conventions they were first run with, including every surrogate p-value.

Two quantities in earlier internal result files differ from the values stated here. The synchronization–amplitude correlation of early drafts, +0.51, applied the Hilbert transform to the 1970–2022 window only after a 1950–2022 band-pass (convention W of §3.6.2); it reproduces exactly under that convention. The correlation between mean pairwise cycle correlation and the order parameter quoted in early drafts did not reproduce from the raw data; under the conventions stated here it is +0.50 on the 26 economies with equal weights (+0.51 on the 31). Neither changes a conclusion.

Figure — Methods diagnostics (F5.4, supplementary). Panel **a**: MST mean edge length by year from 1971 (blue) with the decade means for 1979–89 and 2013–22 (orange), 0.687 against 0.424, a factor of 1.6×. Panel **b**: the phase-lag spread against its 5,000-member surrogate null — the observed 0.546 yr lies below every surrogate ($p = 0.0002$, the floor). Panel **c**: raw (filled) and Bonferroni-corrected (open) p-values for the five lags of §3.5.3, against the $1/53$ permutation floor (dotted) and 0.05 (dashed).

4 Results

Synchronization among the 31 economies is far from constant (Figure 1, F1.3). The order parameter ranges from 0.01 (1985) to 0.95 (2019) and averages 0.60 in global-recession years, 0.48 in normal years and 0.30 in the milder global downturns of 1998, 2001 and 2012 (descriptive; 95% bootstrap intervals in Appendix Table A6). In the two most recent recessions nearly every economy contracted together: 25 of the 31 in 2009 and 28 in 2020, against 9 in 1982 and 9 in 1991 (Global Macro Database real GDP). Five results follow; every other test, including those that came out null, is reported in the Methods (§3.5–§3.6) and the Appendix.

4.1 Synchronization and swing size move together

When the economies' cycles are more aligned, they are also larger. After removing slow trends, the correlation between $r(t)$ and the mean swing size $\bar{A}(t)$ is +0.646 over 1970–2022 (Figure 2, F4.1). No circular shift of one series against the other does better ($p = 0.019$, the floor), and no one of 5,000 sets of spectrum-preserving surrogate economies does either ($p = 0.0002$, the floor). The link holds on the 30 (+0.641) and the original 26 (+0.608), without Iran and Saudi Arabia, whose swings are the largest (+0.625), and under every phase convention, including phases from past data only (+0.726); its size depends on the convention (§3.6.2). This test uses no recession dates.

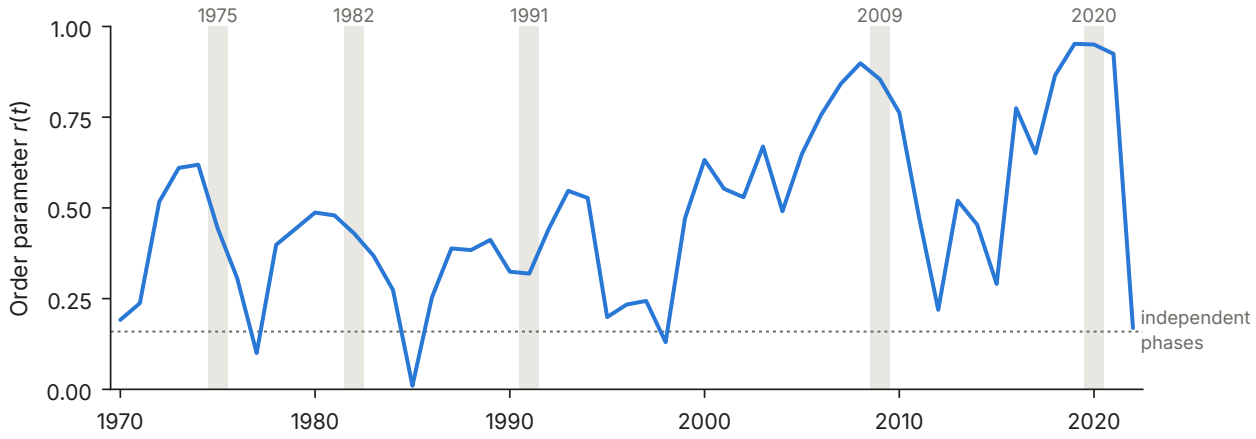


Figure 1 (F1.3). The Kuramoto order parameter $r(t)$ of the 31 economies, 1970–2022, convention P. Grey bands: the five global recessions. Dotted line: the mean of r under independent random phases.

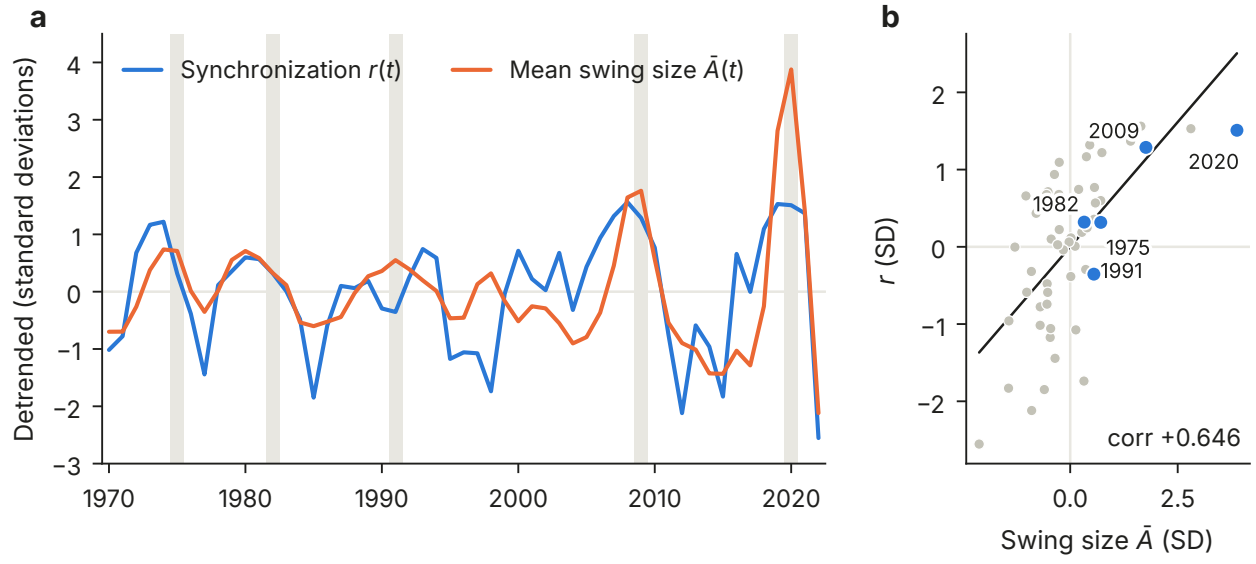


Figure 2 (F4.1). Synchronization and swing size. (a) $r(t)$ and the mean swing size $\bar{A}(t)$, each minus a Gaussian-smoothed trend ($\sigma = 8$ years) and standardized; grey bands: global recessions. (b) The same 53 years as pairs, recession years in blue, with the least-squares line.

4.2 Recession variance is mostly co-movement

In global recessions the world economy loses diversification: the economies' cycles stop offsetting one another. On rolling 10-year windows the variance of the GDP-weighted world cycle splits exactly into an own term (each economy's volatility) and a cross term (co-movement). The cross term supplies 84.3% of the extra world variance in recession windows (97.0% with equal weights; on the 30, 83.3% and 96.8%; Figure 3, F4.4). This is a descriptive magnitude, not a significance claim: the share has no proper null (§3.6.6), and the cross term dominates world variance in any period because 465 pairs outnumber 31 variances (§3.5.1). What is specific to recessions is the gap: the cross term's share of world variance is 0.706 in recession windows against 0.620 in normal ones.

The cross term rises for two reasons, and they are of similar size. Raising only the pairwise correlations from their normal to their recession means supplies 46% of its rise; raising only the swing sizes supplies 40%; their interaction the remaining 14%. We read this as **amplification transmitted by alignment**: larger swings matter for world output because they are aligned, and which of the two parts is larger depends on the sample (on the original 26, swing size 53% against correlation 33%).

The same arithmetic can be written for a single year (Box 1, §5.2). If a common shock reaches each economy according to where it is in its cycle, the world response has the variance of $N_{\text{eff}} = 1/r^2$ independent economies: with the phases spread out the 31 economies behave like 31, with the phases aligned they behave like one. Over 1970–2022, $N_{\text{eff}} = 1/E[r^2]$ is about 3.5 on average, 2.4 in recession years and 2.1 in the year before, and 1.1 in 2020 (descriptive; Appendix Table A7). The dip is carried by 2009 (1.37) and 2020 (1.11); the earlier recession years are not low (1975 5.1, 1982 5.4, 1991 9.8). Averaged over the decomposition's windows the variance law gives the cross term 84.1% of the extra recession variance, against the 84.3% measured: the loss of diversification is this arithmetic, and it is an accounting of the recession excess, not a cause of it.

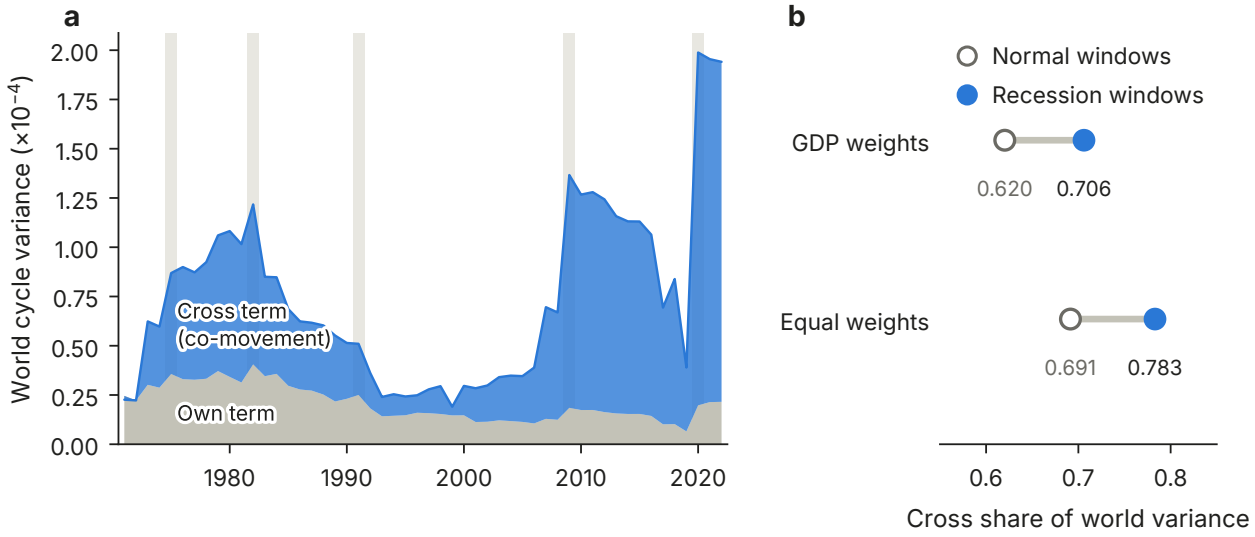


Figure 3 (F4.4). Loss of diversification. (a) Variance of the GDP-weighted world cycle on 10-year windows, split into the own term (grey) and the cross term (blue); grey bands: windows ending in a recession year. (b) The cross term's share of world variance in normal and recession windows, with GDP and with equal weights. No p-value is shown: the share has no proper null (§3.6.6).

4.3 The network has tightened

The correlation network of the 31 business cycles has tightened over five decades. The mean edge length of its minimum spanning tree fell from 0.687 in 1979–89 to 0.424 in 2013–22, a factor of 1.6× (Figure 4, F5.1; 1.6× on the 30 and 1.8× on the original 26). This is a description of a trend, not a test. The network does not contract in the years before recessions ($\tau = -0.157$, $p = 0.206$; §3.5.5),

and the economies move too nearly together for a leader to be identified: the spread of their phase leads is 0.546 years, below every one of 5,000 surrogates ($p = 0.0002$, the floor; §3.5.6).

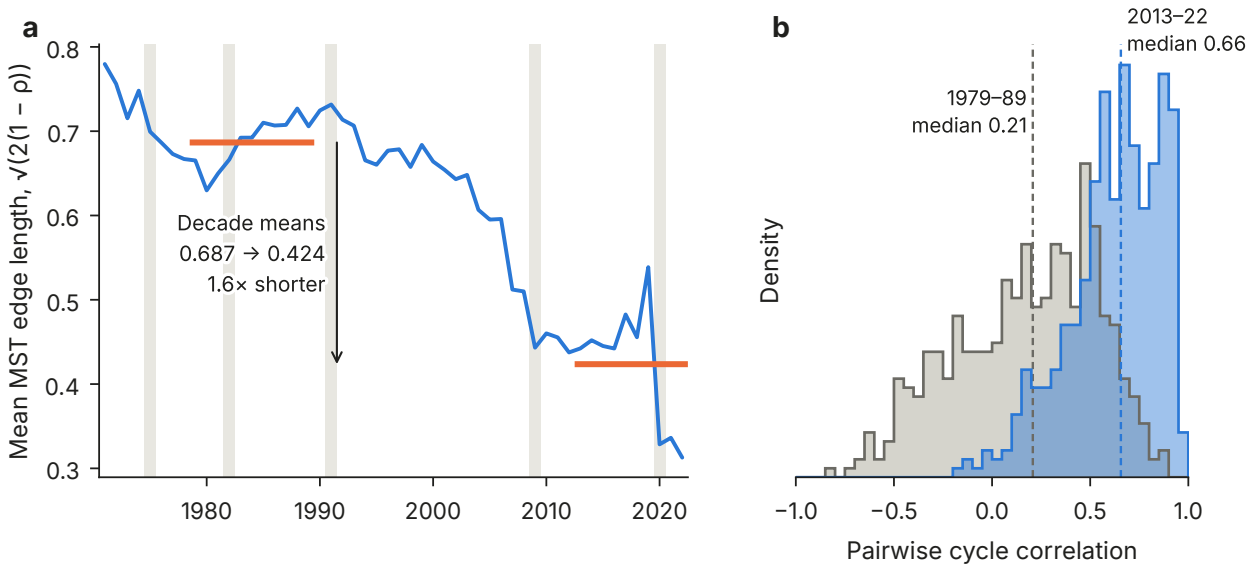


Figure 4 (F5.1). The network has tightened. (a) Mean edge length of the minimum spanning tree of the 31 business cycles on 10-year windows, with the decade means for 1979–89 and 2013–22 (orange). (b) The 465 pairwise cycle correlations, averaged over the windows ending in each decade; dashed lines: medians.

4.4 Synchronization rises in the year before global recessions

With phases built from past data only — the version that uses no information from after each year — synchronization is higher in the year before global recessions: the lead-1 correlation is +0.379 ($p = 0.038$ against 53 circular shifts, whose floor is 0.019), **and it rests on two events, 2009 and 2020** ($n \approx 2$): without either the two-sided lead is not significant ($p = 0.226$ and 0.283), and without both it is gone (−0.02). With China left out the past-only lead-1 is +0.357 ($p = 0.057$). With the usual two-sided phases the lead-1 correlation is +0.231, at the floor of the circular-shift null ($p = 0.019$) and $p = 0.033$ against surrogates with ends (§3.4.4); it survives the removal of any one of the 31 economies ($p \leq 0.038$; Appendix Table A3) and the edge-robust convention ($p = 0.019$), but not convention N ($p = 0.057$), and it does not pass a correction for testing five lags, which with 53 shifts a single lag cannot do (§3.6.4). We lead with the past-only value because the two-sided arc is inflated by how recessions are dated (Limitations). On the 30 the past-only lead-1 is +0.369 ($p = 0.038$) and on the 26 +0.374 ($p = 0.038$).

The rest of the arc is not claimed (Figure 5, F2.5). The two-year lead (lead-2 +0.235, $p = 0.038$) does not survive past-only phases (−0.044, $p = 0.585$). The crisis year is not significant with two-sided phases ($p = 0.189$); with past-only phases it is at the floor (+0.417, $p = 0.019$), which was not a pre-registered claim and is reported, not interpreted. The fall two years after recessions (−0.258, $p = 0.057$) is in the Limitations. This is a property of the tightly coupled modern economy, observed twice; it is not a fifty-year regularity and not a forecast.

4.5 No critical slowing down

Before recessions the variance of synchronization rises ($\tau = +0.438$, $p = 0.031$ against 5,000 surrogates; $p = 0.029$ against surrogates with ends) while its lag-1 autocorrelation does not ($\tau = -0.152$, $p = 0.789$), and the recovery rate does not fall after recessions ($p = 0.502$; Figure 6, F6.1). Critical slowing down — the signature of a system approaching a bifurcation — raises variance and autocorrelation together, because both come from a slower return to equilibrium. A rise in variance alone is not that signature. *As an interpretation, not a test:* it is what noise-induced tipping would

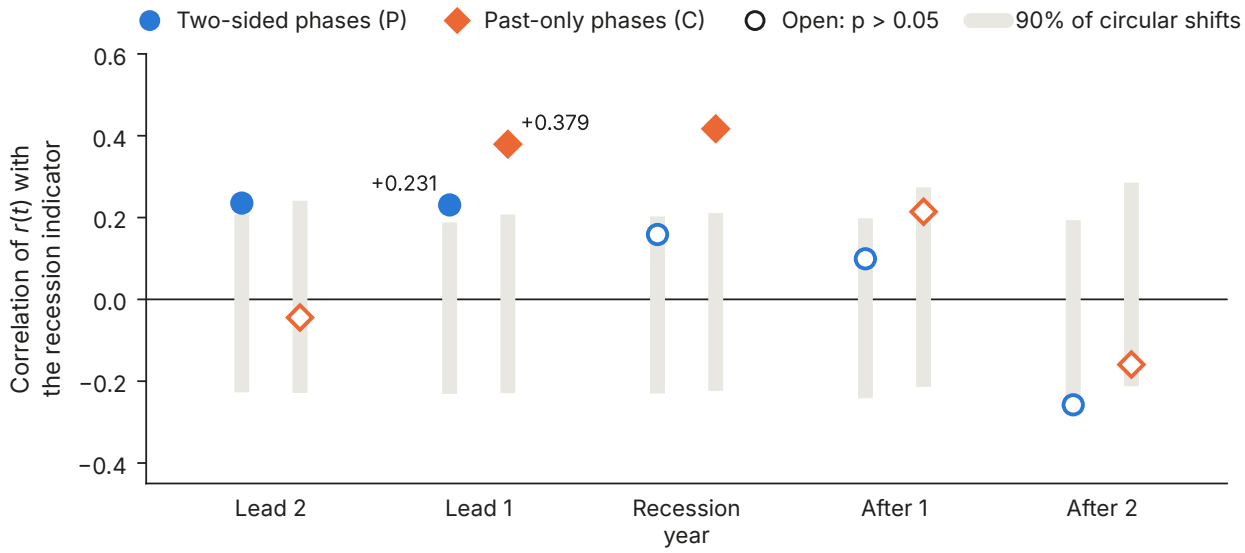


Figure 5 (F2.5). The tested arc. Correlation of $r(t)$ with the recession indicator at each lag, with two-sided phases (P, circles) and phases from past data only (C, diamonds). Filled: one-sided $p \leq 0.05$ against 53 circular shifts (floor 0.019); grey bars: the range of 90% of the shifted correlations.

show, where larger disturbances push the system across a threshold while its restoring forces stay the same (Ashwin et al. 2012). The variance rise, like the lead, rests on 2009 and 2020 (before them alone +0.857, $p = 0.003$; before 1982 and 1991, +0.018, $p = 0.513$; Appendix Table A8), and it is significant under conventions P and E but not W or N (§3.5.7).

5 Discussion

5.1 What the transition is, and what it is not

We first asked whether global recessions arrive the way a system approaches a bifurcation, with recovery slowing as a threshold nears. They do not: before recessions the autocorrelation of synchronization does not rise, and after them the recovery rate does not fall (§4.5). That "what it isn't" led to the account the paper proposes: loss of diversification. In recessions the economies' cycles stop offsetting one another, and co-movement supplies most of the extra world variance (§4.2). Because co-movement and larger swings contribute about equally, we read the recession excess as amplification transmitted by alignment: shocks that would have cancelled across economies are passed along instead. The rise in the variance of synchronization before recessions, without a rise in its autocorrelation, fits the same reading — when shocks stop cancelling, synchronization swings more widely without recovering more slowly — and in the taxonomy of Ashwin et al. (2012) it is what noise-induced rather than bifurcation-induced tipping would show. This is a reading of the data, not a separate test, and it rests on 2009 and 2020.

5.2 Box 1: loss of diversification as arithmetic

Box 1 | Why aligned cycles stop diversifying

Let economy i's cycle have phase θ_i and swing size A_i, and let a shock reach every economy according to where it is in its cycle: economy i responds with $q_i = \sigma A_i \, u \cdot v_i$, where $v_i = (\cos\theta_i, \sin\theta_i)$ and u is a unit direction drawn at random. The same disturbance pushes an economy near its peak down and one near its trough up, so economies at opposite points of their cycles cancel in the world average. If only a share ρ of the shock is common and the rest is independent across economies (each with variance $\sigma^2 A_i^2/2$), the variance of the world-

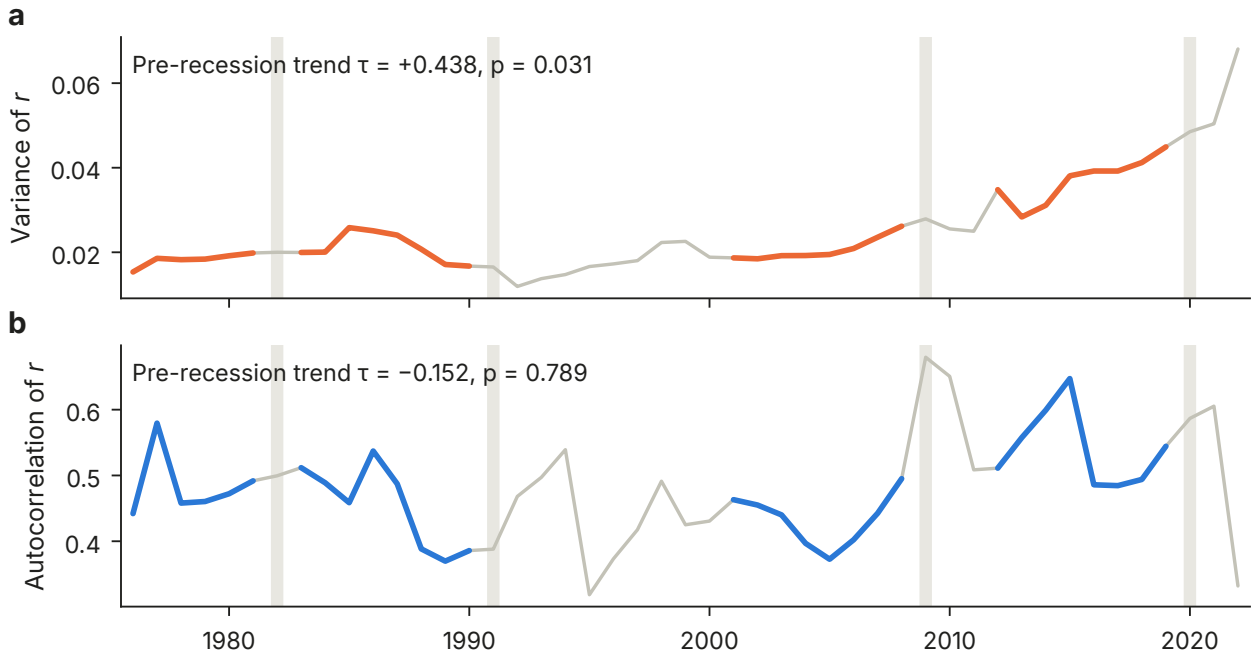


Figure 6 (F6.1). No critical slowing down. Variance (a) and lag-1 autocorrelation (b) of the fluctuations of $r(t)$ on trailing 15-year windows. In colour: the eight years before each recession, over which the trend is tested; grey bands: recession years.

average response $\bar{q}$, given the phases, is

$$\mathrm{Var}(\bar{q}) = \frac{\sigma^2}{2}\left[\rho\,|Z_A|^2 + (1-\rho)\,\frac{\overline{A^2}}{N}\right], \qquad Z_A = \frac{1}{N}\sum_i A_i e^{\mathrm{i}\theta_i}.$$

The first term is the common shock, the second the independent part. Because $|Z_A| = \bar{A}\,R_A$, with $R_A = |\sum_i A_i e^{\mathrm{i}\theta_i}|/\sum_i A_i$ the alignment weighted by swing size, the common term is the square of an exact identity: world swing equals mean swing size times alignment. With equal swing sizes and a fully common shock, $\mathrm{Var}(\bar{q}) = \sigma^2 r^2/2$, against $\sigma^2/(2N)$ if the responses were independent. The world then varies like

$$N_{\mathrm{eff}} = 1/r^2$$

independent economies: the number that would give the same world variance if they did not move together. With a common share ρ, $N_{\mathrm{eff}} = 1/(\rho r^2 + (1-\rho)/N)$. Alignment matters only for the common part of shocks. The independent part is the averaging-out of Lucas (1977), which fails when a few units are large (Gabaix 2011) or central in the input–output network (Acemoglu et al. 2012); here it fails for a third reason: the economies' cycles line up.

On the 31 economies $N_{\mathrm{eff}} = 1/E[r^2]$ is about 3.5 over 1970–2022, 2.4 in recession years and 2.1 the year before (1.1 in 2020); 2009 and 2020 pull these down, and the earlier recessions are not low. Averaged over the 10-year windows of §4.2, the law gives the cross term 84.1% of the extra world variance in recession windows, against 84.3% measured.

It is arithmetic, not a mechanism. Any year in which cycles happen to line up has a larger world swing, coupled or not, and recessions dated from world output are such years even in simulated worlds with no coupling (§5.4).

Interpretation, not a test: in the taxonomy of Ashwin et al. (2012), a transition caused by larger disturbances while the restoring forces stay the same is noise-induced tipping, which does not show critical slowing down — and the autocorrelation of synchronization does not rise before recessions, nor does recovery slow after them (§4.5).

The law is exact for the stated shock, which we check against simulation (the code is in the repository); its link to the decomposition is exact year by year (with equal weights and swing sizes, the cross share is $1 - N_{\text{eff}}/N$) and close, not exact, over the decomposition's windows. With GDP weights the independent term becomes $(1-\rho)\sum_i w_i^2 A_i^2/2$, a Herfindahl-type concentration — Gabaix's granular channel — and the common term $\rho\,|\sum_i w_i A_i e^{\mathrm{i}\theta_i}|^2/2$ is the alignment channel. The two are complements, not rivals.

5.3 Diversification-type tipping

We call a transition of this kind diversification-type tipping: a transition caused by losing the cancellation between economies rather than by slowing recovery. It differs from bifurcation-induced tipping, which shows the slowing-down signature we do not find. In finance, "diversification" and systemic risk usually describe a different mechanism — institutions that each diversify become alike and more connected, so diversification itself can spread distress (Burkholz, Garas and Schweitzer 2016) — and our sense is narrower: cycles that stop offsetting one another. This paper does not show that loss of diversification has a threshold, a level of coupling beyond which cancellation fails abruptly rather than gradually.

5.4 The mechanism check: simulated worlds with known coupling

Two simulation studies, each pre-registered before it was run, ask what the tests can and cannot show.

G1: worlds with known coupling. We simulated 31 economies as phase oscillators (Kuramoto) and as phase-and-size oscillators (Stuart–Landau) on the real trade network, calibrated to the real cycles, and ran every test of this paper on 1,000 worlds per setting. With no coupling the tests reject at about their stated rates (Table 1, §3.4.3). Two findings bear on the claims. First, when recessions are dated from each simulated world's own output, as global recessions are, the two-sided crisis-year and lead-1 arcs reject in 8.2%–8.9% and 8.5%–8.6% of uncoupled worlds, about twice the correct rate, while the past-only lead-1 does not (2.3%–2.9%): this is why the lead claim rests on the past-only version. Second, no G1 world produced the past-only lead: with own-dated recessions it rejected in 1.7% of trade-coupled, 2.9% of uncoupled and 2.5% of commonly forced Kuramoto worlds, never above chance.

G1b: a synchrony-driven escape layer. We then added the mechanism Box 1 suggests to the same Kuramoto worlds: a recession is an escape of a world-level state from a stable well, $dx = (\mu + x - x^3)\,dt + \sigma\, r(t)\, dW$, so that shocks grow with the worlds' true synchrony. σ was set so that worlds have five recession years in 53 on average, using only their number, never their timing; each escape resets the state and starts a two-year refractory period. Before running we fixed what "reproduces the lead" means: (a) the past-only lead-1 rejects more often than chance (the lower bound of its 95% interval above 3.8%); (b) more often than the same layer with constant noise and with synchrony shuffled in time (both paired differences' intervals above zero); and (c) at least 20% of worlds reach the real $p = 0.038$. Some lead is expected by construction, since the layer makes recessions likelier when synchrony is high; the question is how much survives annual sampling, 53 points, five events and past-only phases.

The verdict, exactly as pre-registered, is **partly** (Appendix Table A11). In trade-coupled worlds the past-only lead-1 rejects in 6.3% of worlds (95% interval 5.0%–8.0%), above chance (a) and above both controls (b: 3.5 percentage points more than with constant noise, 3.4 percentage points more than with shuffled synchrony). But only 6.3% of worlds reach the real p, against the 20% required (c fails), and 1.3% reach the real correlation of +0.379. Without trade coupling the layer leaves no past-only lead (4.0% with no coupling, 3.2% with common forcing only). Of eight pre-registered sensitivities on the trade-coupled worlds, 6 give the same verdict and 2 do not reproduce the lead (one state per economy; refractory 4 years); none reaches (c).

What this says, and no more: a synchrony-driven escape mechanism on coupled economies can leave a past-only lead that our pipeline detects, and coupling is needed for it, but a lead as large as

the real one is rare under it. Either the real lead is an unusual draw — which, resting on two events, it may well be — or the real mechanism is stronger than, or different from, the one simulated; the simulated recessions, for instance, do not feed back on the cycles they are measured from. We claim no mechanism from this check.

5.5 Where this sits

The world business-cycle factor of Kose, Otrok and Whiteman (2003) measures how much economies move together; we ask when they do, and whether togetherness itself is the danger. Treating national cycles as coupled oscillators follows Selover and Jensen (1999), who described mode-locking of business cycles, and Ikeda, Aoyama and Yoshikawa (2013), who fitted Kuramoto-type couplings between six economies' quarterly cycles; Pangallo (2025) shows that trade can synchronize endogenous cycles across 24 countries. Morikawa, Morikawa and Nakamichi (2026) use the same Hilbert and Kuramoto toolkit for a different question, the 1/f spectra of economic series. None of these relates synchronization to the timing of global recessions or to the loss of diversification. We do not forecast: two-sided filters cannot support a forecast, and the past-only lead rests on two events.

6 Limitations

Two events carry the lead and the variance rise. The pre-recession rise rests on 2009 and 2020 ($n \approx 2$): without either the two-sided lead-1 is not significant ($p = 0.226$ and 0.283), the split sample 1970–1995 shows none (lead-1 +0.24, $p = 0.154$), and the variance rise of §4.5 has the same dependence. Both are properties of the tightly coupled modern economy observed twice, not regularities, and with five events in 53 years no test here has more than five degrees of freedom.

What multiplicity correction can show. No lag survives a correction for testing five lags. With 53 circular shifts the smallest p the test can return is 0.019, so a single lag can never pass Bonferroni and passes Benjamini–Hochberg only alongside another lag at the floor, which happened on the original 26 but not on the 30 or the 31 (Appendix Table A5). The pre-crisis rise therefore rests on its survival under past-only phases and leave-one-economy-out, not on corrected significance.

How global recessions are dated. Global recessions are dated from world output, and the economies whose synchronization we measure make up most of it. The years in which world output falls most are years in which many cycles happen to be falling together, so dating recessions this way builds some synchronization into them even when economies are not coupled at all. In simulated worlds with no coupling, recession years dated the same way from each world's own output make the two-sided arc reject at the crisis year in 8.2%–8.9% and at lead-1 in 8.5%–8.6% of worlds, about twice the correct 3.8% (§3.4.3). Phases built from past data only are free of this (2.3%–2.9%). The lead claim therefore rests on the past-only lead-1 (+0.379, $p = 0.038$), not on the two-sided one.

What the cross-term test shows. The surrogate test of the cross term (§3.6.6) asks whether the cross term's share of world variance in recession windows exceeds what independent economies would give. In simulated worlds it rejects in nearly every coupled world whatever dates are used — 97.1%–99.3% of trade-coupled worlds with the real recession dates, which have nothing to do with those worlds — and in about 5% of uncoupled ones. It detects co-movement in general, not co-movement at recessions. The recession-specific claim — that co-movement supplies more of world variance in recessions than in normal years — rests on the gap between recession and normal windows (cross share 0.706 against 0.620, GDP-weighted), which is descriptive.

The end of the sample. 2020 sits two years from the last observation, and both the forward–backward filter and the Hilbert transform are least reliable there. Without 2020 the two-sided lead-1 is +0.11 ($p = 0.283$). Surrogates with ends (§3.4.4) move the two-sided lead-1 from 0.019 to 0.033 and leave the claimed results on the same side of 0.05, but the most recent event remains the least well measured; data for 2023–2025 are the cleanest fix.

The fall after recessions. Synchronization falls in the two years after a global recession (lag -2:

-0.258 on the 31, $p = 0.057$; -0.236, $p = 0.057$ on the 30; -0.277, $p = 0.019$ on the original 26), but this half of the arc does not hold up. It is not significant on the 31 or the 30; it disappears with the 3–10 year band and with growth rates (§3.6.1–3.6.2); it is significant under none of the alternative phase conventions, including phases from past data only; and without 2020 it is -0.18 ($p = 0.151$). On the 31 it sits on the threshold: leaving out any of 20 of the 31 economies, or computing Russia's phases from 1992, takes it below 0.05. It also sits where the filters are least reliable: the lag -2 test for 2020 uses $r(2022)$, the last observation. Against surrogates with ends it is $p = 0.039$ on the 31 and 0.054 on the 30; that null still rejects lag -2 slightly too often in simulated coupled worlds (6.5%), and most of the move is the finer resolution of 5,000 surrogates against 53 shifts. We report it and do not claim it.

Phase conventions. The arc depends on where the Hilbert transform is applied (Table 2). Lead-1 is significant under P, W, E and C but not N ($p = 0.057$), and on the 30 not under W ($p = 0.057$); the variance rise is significant under P and E but not W or N. The synchronization–swing-size link holds under all five, with sizes from +0.547 to +0.726. We chose P before running anything on the added economies, on the ground that its run-in protects 1970 from edge transients, not because it gives larger values.

The dropped rate lead. With two-sided phases short-term interest rates are most synchronized two years before global recessions (+0.273, $p = 0.019$, 19 economies with continuous rates), but with phases from past data only the rise is gone (+0.165, $p = 0.151$). We make no claim that rates or investment signal recessions in advance; the channel ordering is description only (Appendix Table A10).

The alignment share is sample-sensitive. Splitting the recession excess in world swing (the identity of Box 1) into alignment and swing size gives an alignment share of 28% on the 31, 19% on the 30, 33% on the 30 without Iran and 39% on the original 26. The swing-size part is significant on every sample ($p = 0.0002$, the floor, also against surrogates with ends); the alignment part on none ($p = 0.239$ on the 31). The share depends on which economies are in the sample, and we draw no conclusion from it.

The mechanism check is narrow. G1b tested one mechanism in one family of simulated worlds (Kuramoto oscillators on the real 1970–2022 trade network), with recessions that do not feed back on the cycles and a calibration that fixes only how often recessions happen. Its *partly* verdict says that this mechanism can leave a detectable past-only lead, not that it produces the real one.

Data. Annual data give 53 points; Maddison's PPP series for Russia before 1991 are back-cast Soviet-era estimates (every test is repeated with Russia's phases from 1992); Maddison's 1991–2011 growth for Norway, Switzerland and South Africa runs above national accounts (replacing it changes no conclusion; §3.6.8). The five recession dates are those of Kose, Sugawara and Terrones (2020) for 1975–2009 and of the World Bank (2020) for 2020, under the same definition.

7 Future work

Paper 1 is closed to new tests. What follows is the programme for a second paper, which has not been started.

A graph neural network of coupled economies (stages G2–G4). A first stage built yearly graphs of the 31 economies with pinned trade, bank-lending and oil links and a world node (G0), and a simulator of coupled economies with known coupling run through this paper's pipeline (G1, used in §5.4). The next stages are: G2, a recoverability benchmark that asks which couplings 53 annual points can reveal at all, and by which method; G3, a model of the real network, trained only after G2 and checked against a twin with no network, shuffled edges, leave-one-recession-out validation, past-only phases and classical baselines (a trade-weighted global VAR, a Kuramoto regression, a network autoregression); and G4, explanations of what the model learned, counted only after the same explanation method recovers known links in the simulator. The aim is to understand which links carry synchronization, not to forecast recessions.

Channel experiments. The channels inside each economy (interest rates, investment, trade,

budgets) synchronize at different times around recessions with two-sided phases, but no channel lead survives past-only phases. Testing the ordering properly needs the channel data to be extended and the past-only design to be fixed in advance for each channel.

The remaining DTT modules. The diversification-type tipping report (Sid, 13 October 2026) has modules this paper does not use: cascades between economies, asymmetric restoring forces and hysteresis. Each needs its own pre-registration and a check in the simulator, where the coupling can be raised step by step to test whether loss of diversification has a threshold. Rate-induced tipping, where the speed of a change rather than its size causes the transition, is a further hypothesis for that work.

Data for 2023–2025. The most recent recession sits two years from the end of the sample, where the filters are least reliable. Extending the series by three years, when Maddison's next release covers them, is the cleanest test of the 2020 result and of the fall after recessions.

A quarterly track. Quarterly national accounts would give four times as many points but start around 1981 and lose the 1975 recession; a quarterly version is a test of whether the annual results survive finer timing, not a replacement.

Banking-crisis dating. The dates of systemic banking crises (the Jordà–Schularick–Taylor database), tried early in this project, were not re-verified on pinned data; no claim here depends on them.

AI-assistance disclosure

This paper was written with substantial help from AI tools, and we say so plainly. Claude (Anthropic) was used for analysis, code and drafting: it wrote and ran most of the code in the repository that accompanies the paper (the data build, the pipeline, the tests, the simulations and the figures), drafted text that the author revised, and checked references. The report on diversification-type tipping that Box 1 draws on (Sid Shinde, 13 October 2026) was drafted with an OpenAI tool; its equations were re-implemented and checked independently before any of them was used here. Every number in the paper is produced by code in the repository from pinned data, and every hypothesis, test direction and sample was written down and committed before the tests were run. The research questions, the choices among alternatives and the claims are the author's, and the author is responsible for any errors.

Data and code availability

The code, the pinned data manifests, the pre-registrations, every results file and the figures are in the public repository `https://github.com/fxcl1618033/economic-fibrillation`; the version described here is the tag `paper1-complete-v1.3`. Running `make reproduce` regenerates every number in the paper from the pinned data and compares each file with the committed one; `make all` is a quicker check. The data come from the Maddison Project Database 2023, the Global Macro Database, Penn World Table 10.01, the IMF, the BIS, the EIA and the World Bank, each pinned to a repository commit or a dated download with its checksum (`data/sources.yaml`). Series from the Global Macro Database are not redistributed, as its terms require; the repository rebuilds them from the pinned release. The code is under the MIT licence and the text, figures and results under CC BY 4.0, except the outputs built on the Global Macro Database, which follow its terms (CC BY-NC-SA 4.0).

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Appendix

All values are convention P unless a row says otherwise; one-sided p in brackets. Circular-shift p-values have a floor of $1/53 = 0.019$ (for the split samples of 26 and 27 years, 0.038 and 0.037); surrogate p-values with 5,000 draws have a floor of 0.0002, with 2,000 draws 0.0005.

Table A1. Key results by sample

The original 26 economies, the 30 (without Russia), the 31, the 31 without each added economy, with Russia's phases from 1992 only and without Iran and Saudi Arabia together. "—": not run (the variant without Iran and Saudi Arabia was pre-registered for the swing-size tests only, the one with Russia from 1992 for the circular-shift tests only).

Quantity	S26	S30	S31	S31 without China	without Brazil	without Saudi Arabia	without Iran	without Russia (= S30)	Russia from 1992	without Iran and Saudi Arabia
Arc, lead-2	+0.230 (0.038)	+0.226 (0.038)	+0.235 (0.038)	+0.226 (0.057)	+0.225 (0.038)	+0.252 (0.019)	+0.244 (0.038)	+0.226 (0.038)	+0.229 (0.038)	—
Arc, lead-1	+0.216 (0.019)	+0.217 (0.019)	+0.231 (0.019)	+0.231 (0.019)	+0.224 (0.019)	+0.229 (0.038)	+0.241 (0.019)	+0.217 (0.019)	+0.224 (0.019)	—
Lead-1, past-only phases	+0.374 (0.038)	+0.369 (0.038)	+0.379 (0.038)	+0.357 (0.057)	+0.391 (0.038)	+0.403 (0.038)	+0.373 (0.038)	+0.369 (0.038)	—	—
Arc, after-2 (lower tail)	−0.277 (0.019)	−0.236 (0.057)	−0.258 (0.057)	−0.262 (0.038)	−0.274 (0.019)	−0.260 (0.038)	−0.274 (0.019)	−0.236 (0.057)	−0.254 (0.038)	—
r with swing size, detrended	+0.608 (0.019; surr. 0.0002)	+0.641 (0.019; surr. 0.0002)	+0.646 (0.019; surr. 0.0002)	+0.656 (0.019; surr. 0.0002)	+0.628 (0.019; surr. 0.0002)	+0.625 (0.019; surr. 0.0002)	+0.650 (0.019; surr. 0.0002)	+0.641 (0.019; surr. 0.0002)	+0.655 (0.019)	+0.625 (0.019; surr. 0.0002)
Cross share of the recession excess, GDP weights	80.2%	83.3%	84.3%	83.1%	83.4%	83.8%	84.0%	83.3%	—	83.5%
Same, equal weights	96.5%	96.8%	97.0%	96.9%	96.8%	97.3%	96.7%	96.8%	—	97.0%
Cross-term rise: correlation / swing size / interaction	33 / 53 / 14%	44 / 41 / 15%	46 / 40 / 14%	41 / 45 / 14%	45 / 42 / 14%	45 / 41 / 14%	44 / 42 / 14%	44 / 41 / 15%	—	43 / 43 / 14%
Alignment share of the swing excess (p)	39.3% (0.103)	18.9% (0.356)	28.5% (0.239)	32.6% (0.179)	24.7% (0.276)	28.6% (0.242)	38.3% (0.119)	18.9% (0.356)	—	41.4% (0.095)
MST decade means, 1979–89 vs 2013–22	0.677 vs 0.379 (1.79×)	0.680 vs 0.421 (1.62×)	0.687 vs 0.424 (1.62×)	0.682 vs 0.418 (1.63×)	0.694 vs 0.414 (1.67×)	0.685 vs 0.416 (1.65×)	0.684 vs 0.412 (1.66×)	0.680 vs 0.421 (1.62×)	—	—
Phase-lead spread (lower-tail p)	0.445 yr (0.0002)	0.535 yr (0.0004)	0.546 yr (0.0002)	0.535 yr (0.0006)	0.542 yr (0.0004)	0.545 yr (0.0006)	0.474 yr (0.0002)	0.535 yr (0.0004)	—	—
Autocorrelation trend before recessions	+0.027 (0.456)	−0.098 (0.676)	−0.152 (0.789)	−0.205 (0.839)	−0.098 (0.684)	+0.009 (0.487)	−0.152 (0.764)	−0.098 (0.676)	—	—
Variance trend before recessions	+0.438 (0.033)	+0.420 (0.037)	+0.438 (0.031)	+0.438 (0.033)	+0.402 (0.045)	+0.438 (0.031)	+0.420 (0.037)	+0.420 (0.037)	—	—
Recovery-rate change after recessions	+0.078 (0.534)	+0.024 (0.507)	+0.002 (0.502)	−0.022 (0.480)	−0.017 (0.472)	+0.027 (0.498)	+0.068 (0.517)	+0.024 (0.507)	—	—
Short-rate synchronization, lead-2 [economies]	+0.266 (0.019) [18]	+0.273 (0.019) [19]	+0.273 (0.019) [19]	+0.273 (0.019) [19]	+0.266 (0.019) [18]	+0.273 (0.019) [19]	+0.273 (0.019) [19]	+0.273 (0.019) [19]	—	—

Table A2. Robustness of the arc

Correlation of $r(t)$ with the recession indicator at each lag (one-sided p; after-2 tested downward). 2–9 year band unless stated.

The 31 economies.

Check	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
Baseline	+0.24 (0.038)	+0.23 (0.019)	+0.16 (0.189)	−0.26 (0.057)
Band 2–8 years	+0.23 (0.057)	+0.19 (0.094)	+0.12 (0.226)	−0.27 (0.019)
Band 3–10 years	+0.26 (0.038)	+0.25 (0.019)	+0.16 (0.132)	+0.00 (0.509)
HP filter, $\lambda = 100$	+0.27 (0.038)	+0.29 (0.019)	+0.14 (0.113)	−0.18 (0.094)
Growth rates instead of band-pass	+0.25 (0.057)	+0.34 (0.038)	+0.15 (0.132)	+0.04 (0.679)
1970–1995 only (1975, 1982, 1991)	+0.34 (0.077)	+0.24 (0.154)	+0.06 (0.423)	−0.21 (0.192)
1996–2022 only (2009, 2020)	+0.32 (0.111)	+0.38 (0.037)	+0.35 (0.074)	−0.36 (0.074)
Without 1975	+0.22 (0.057)	+0.22 (0.019)	+0.19 (0.113)	−0.16 (0.151)
Without 1982	+0.26 (0.038)	+0.26 (0.019)	+0.19 (0.075)	−0.22 (0.038)
Without 1991	+0.28 (0.038)	+0.31 (0.019)	+0.23 (0.057)	−0.30 (0.038)
Without 2009	+0.14 (0.208)	+0.13 (0.226)	+0.06 (0.358)	−0.28 (0.057)
Without 2020	+0.14 (0.208)	+0.11 (0.283)	+0.03 (0.434)	−0.18 (0.151)

Check	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
Without 2009 and 2020	+0.02 (0.377)	−0.02 (0.491)	−0.09 (0.717)	−0.20 (0.094)
Each of the 31 economies left out (range)	+0.22 to +0.26 (0.019–0.075)	+0.22 to +0.25 (0.019–0.038)	+0.13 to +0.17 (0.170–0.208)	−0.27 to −0.24 (0.019–0.075)
Norway, Switzerland, South Africa with GMD growth 1991–2011	+0.24 (0.057)	+0.24 (0.019)	+0.16 (0.170)	−0.28 (0.038)

The 30 economies (without Russia).

Check	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
Baseline	+0.23 (0.038)	+0.22 (0.019)	+0.16 (0.189)	−0.24 (0.057)
Band 2–8 years	+0.22 (0.038)	+0.18 (0.113)	+0.12 (0.226)	−0.26 (0.038)
Band 3–10 years	+0.26 (0.038)	+0.24 (0.019)	+0.16 (0.113)	+0.02 (0.566)
HP filter, $\lambda = 100$	+0.28 (0.038)	+0.27 (0.019)	+0.14 (0.113)	−0.19 (0.075)
Growth rates instead of band-pass	+0.24 (0.075)	+0.33 (0.038)	+0.14 (0.132)	+0.04 (0.679)
1970–1995 only (1975, 1982, 1991)	+0.31 (0.077)	+0.21 (0.154)	+0.06 (0.385)	−0.18 (0.192)
1996–2022 only (2009, 2020)	+0.32 (0.111)	+0.37 (0.037)	+0.35 (0.074)	−0.34 (0.074)
Without 1975	+0.21 (0.057)	+0.20 (0.057)	+0.18 (0.132)	−0.15 (0.151)
Without 1982	+0.25 (0.038)	+0.25 (0.019)	+0.20 (0.057)	−0.19 (0.038)
Without 1991	+0.28 (0.038)	+0.29 (0.019)	+0.24 (0.057)	−0.28 (0.038)
Without 2009	+0.14 (0.208)	+0.12 (0.226)	+0.07 (0.358)	−0.26 (0.075)
Without 2020	+0.12 (0.264)	+0.10 (0.321)	+0.03 (0.453)	−0.17 (0.170)
Without 2009 and 2020	+0.01 (0.377)	−0.03 (0.528)	−0.09 (0.679)	−0.19 (0.113)
Each of the 30 economies left out (range)	+0.21 to +0.25 (0.019–0.094)	+0.21 to +0.23 (0.019–0.038)	+0.13 to +0.18 (0.132–0.226)	−0.25 to −0.22 (0.019–0.075)
Norway, Switzerland, South Africa with GMD growth 1991–2011	+0.23 (0.038)	+0.22 (0.019)	+0.16 (0.151)	−0.26 (0.075)

The original 26 economies.

Check	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
Baseline	+0.23 (0.038)	+0.22 (0.019)	+0.15 (0.189)	−0.28 (0.019)
Band 2–8 years	+0.22 (0.038)	+0.19 (0.057)	+0.12 (0.245)	−0.31 (0.019)
Band 3–10 years	+0.26 (0.019)	+0.23 (0.038)	+0.16 (0.094)	−0.02 (0.453)
HP filter, $\lambda = 100$	+0.27 (0.038)	+0.27 (0.019)	+0.19 (0.057)	−0.25 (0.038)
Growth rates instead of band-pass	+0.23 (0.094)	+0.34 (0.038)	+0.21 (0.075)	−0.01 (0.509)
1970–1995 only (1975, 1982, 1991)	+0.31 (0.077)	+0.24 (0.115)	+0.06 (0.385)	−0.17 (0.192)
1996–2022 only (2009, 2020)	+0.32 (0.111)	+0.34 (0.037)	+0.32 (0.074)	−0.45 (0.037)
Without 1975	+0.23 (0.038)	+0.21 (0.019)	+0.17 (0.189)	−0.19 (0.113)
Without 1982	+0.25 (0.038)	+0.24 (0.019)	+0.19 (0.057)	−0.23 (0.038)
Without 1991	+0.27 (0.038)	+0.27 (0.019)	+0.21 (0.075)	−0.34 (0.019)
Without 2009	+0.13 (0.226)	+0.12 (0.208)	+0.06 (0.321)	−0.28 (0.057)
Without 2020	+0.14 (0.208)	+0.11 (0.264)	+0.03 (0.434)	−0.19 (0.151)
Without 2009 and 2020	+0.02 (0.377)	−0.01 (0.491)	−0.08 (0.717)	−0.18 (0.094)
Each of the 26 economies left out (range)	+0.21 to +0.26 (0.019–0.113)	+0.20 to +0.23 (0.019–0.057)	+0.12 to +0.17 (0.132–0.264)	−0.30 to −0.26 (0.019–0.038)
Norway, Switzerland, South Africa with GMD growth 1991–2011	+0.23 (0.057)	+0.22 (0.038)	+0.15 (0.170)	−0.30 (0.019)

Table A3. Leave one economy out (31 economies)

Left out	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
United States	+0.228 (0.038)	+0.221 (0.019)	+0.153 (0.189)	−0.250 (0.057)
United Kingdom	+0.231 (0.038)	+0.229 (0.019)	+0.156 (0.170)	−0.258 (0.038)
Germany	+0.237 (0.038)	+0.242 (0.019)	+0.165 (0.189)	−0.274 (0.019)
France	+0.233 (0.038)	+0.234 (0.019)	+0.167 (0.170)	−0.245 (0.075)
Italy	+0.235 (0.038)	+0.228 (0.019)	+0.154 (0.189)	−0.259 (0.038)
Japan	+0.235 (0.038)	+0.232 (0.019)	+0.154 (0.208)	−0.260 (0.038)
Canada	+0.228 (0.038)	+0.226 (0.019)	+0.154 (0.208)	−0.267 (0.038)
Spain	+0.227 (0.038)	+0.227 (0.019)	+0.170 (0.170)	−0.253 (0.038)
Australia	+0.216 (0.075)	+0.222 (0.019)	+0.155 (0.208)	−0.242 (0.075)
South Korea	+0.250 (0.038)	+0.235 (0.019)	+0.146 (0.189)	−0.252 (0.057)

Left out	Lead-2	Lead-1	Crisis year	After-2 (lower tail)
India	+0.245 (0.038)	+0.233 (0.019)	+0.169 (0.170)	−0.267 (0.038)
Mexico	+0.240 (0.038)	+0.242 (0.019)	+0.172 (0.170)	−0.261 (0.038)
Netherlands	+0.235 (0.038)	+0.227 (0.019)	+0.151 (0.208)	−0.258 (0.019)
Sweden	+0.230 (0.038)	+0.233 (0.019)	+0.163 (0.189)	−0.252 (0.057)
Switzerland	+0.232 (0.038)	+0.223 (0.019)	+0.150 (0.189)	−0.264 (0.038)
Belgium	+0.235 (0.038)	+0.229 (0.019)	+0.154 (0.189)	−0.247 (0.075)
South Africa	+0.242 (0.038)	+0.238 (0.019)	+0.165 (0.189)	−0.261 (0.057)
Turkey	+0.261 (0.019)	+0.238 (0.038)	+0.155 (0.189)	−0.262 (0.038)
Norway	+0.254 (0.038)	+0.239 (0.019)	+0.164 (0.170)	−0.271 (0.019)
Austria	+0.231 (0.038)	+0.228 (0.019)	+0.159 (0.189)	−0.249 (0.057)
Denmark	+0.251 (0.038)	+0.247 (0.019)	+0.166 (0.189)	−0.260 (0.038)
Finland	+0.232 (0.038)	+0.231 (0.019)	+0.172 (0.170)	−0.256 (0.075)
Greece	+0.217 (0.057)	+0.229 (0.019)	+0.158 (0.189)	−0.266 (0.019)
Portugal	+0.227 (0.057)	+0.225 (0.019)	+0.168 (0.170)	−0.258 (0.057)
Indonesia	+0.238 (0.038)	+0.221 (0.019)	+0.133 (0.208)	−0.260 (0.038)
Thailand	+0.219 (0.075)	+0.225 (0.019)	+0.145 (0.189)	−0.252 (0.038)
China	+0.226 (0.057)	+0.231 (0.019)	+0.166 (0.189)	−0.262 (0.038)
Brazil	+0.225 (0.038)	+0.224 (0.019)	+0.149 (0.208)	−0.274 (0.019)
Saudi Arabia	+0.252 (0.019)	+0.229 (0.038)	+0.148 (0.189)	−0.260 (0.038)
Iran	+0.244 (0.038)	+0.241 (0.019)	+0.160 (0.189)	−0.274 (0.019)
Russia	+0.226 (0.038)	+0.217 (0.019)	+0.161 (0.189)	−0.236 (0.057)

Table A4. Surrogates with ends

Each test whose statistic uses a year from 2019 to 2022, with its p against the null used in the paper and, after the arrow, against 5,000 surrogates with ends (§3.4.4). **Bold**: the two fall on opposite sides of 0.05. Multivariate surrogates for the arcs, independent ones otherwise.

Test	S31	S30	S26
Arc, lead-1, convention P	+0.231 (0.019 → 0.033)	+0.217 (0.019 → 0.043)	+0.216 (0.019 → 0.042)
Arc, crisis year, convention P	+0.159 (0.189 → 0.116)	+0.161 (0.189 → 0.108)	+0.149 (0.189 → 0.123)
Arc, after-1, convention P	+0.099 (0.717 → 0.760)	+0.111 (0.755 → 0.786)	+0.078 (0.679 → 0.707)
Arc, after-2, convention P	**−0.258 (0.057 → 0.039)**	−0.236 (0.057 → 0.054)	−0.277 (0.019 → 0.026)
Arc within the Fiedler blocs, lead-1	+0.250 (0.019 → 0.023)	**+0.212 (0.019 → 0.050)**	+0.182 (0.057 → 0.082)
Arc between the Fiedler blocs, lead-1	+0.066 (0.358 → 0.354)	+0.135 (0.151 → 0.152)	+0.180 (0.038 → 0.025)
Arc, lead-1, convention W	**+0.204 (0.019 → 0.056)**	+0.184 (0.057 → 0.079)	+0.178 (0.094 → 0.085)
Arc, crisis year, convention W	+0.134 (0.189 → 0.161)	+0.130 (0.226 → 0.167)	+0.112 (0.208 → 0.198)
Arc, after-1, convention W	+0.109 (0.774 → 0.784)	+0.121 (0.774 → 0.806)	+0.088 (0.698 → 0.734)
Arc, after-2, convention W	−0.165 (0.113 → 0.142)	−0.131 (0.170 → 0.204)	−0.144 (0.170 → 0.177)
Arc, lead-1, convention N	+0.192 (0.057 → 0.071)	+0.173 (0.094 → 0.093)	+0.178 (0.094 → 0.083)
Arc, crisis year, convention N	+0.135 (0.208 → 0.156)	+0.133 (0.189 → 0.158)	+0.124 (0.189 → 0.168)
Arc, after-1, convention N	+0.105 (0.755 → 0.775)	+0.116 (0.774 → 0.800)	+0.089 (0.679 → 0.737)
Arc, after-2, convention N	−0.190 (0.094 → 0.101)	−0.158 (0.132 → 0.151)	−0.197 (0.094 → 0.088)
Arc, lead-1, convention E	+0.236 (0.019 → 0.029)	+0.226 (0.019 → 0.037)	+0.220 (0.019 → 0.038)
Arc, crisis year, convention E	+0.159 (0.189 → 0.110)	+0.164 (0.170 → 0.103)	+0.152 (0.170 → 0.118)
Arc, after-1, convention E	+0.101 (0.755 → 0.780)	+0.107 (0.755 → 0.794)	+0.075 (0.717 → 0.719)
Arc, after-2, convention E	−0.160 (0.132 → 0.090)	−0.136 (0.170 → 0.120)	−0.170 (0.113 → 0.080)
Identity: alignment at recessions	+0.406 (0.239 → 0.228)	+0.241 (0.356 → 0.355)	+0.654 (0.103 → 0.105)
Identity: swing size at recessions	+1.021 (0.0002 → 0.0002)	+1.033 (0.0002 → 0.0002)	+1.010 (0.0002 → 0.0002)

Test	S31	S30	S26
r with swing size, detrended (outside the rule)	+0.646 (0.0002 → 0.0002)	+0.641 (0.0002 → 0.0002)	+0.608 (0.0002 → 0.0006)
Cross share at recessions, GDP weights	+0.706 (0.0002 → 0.0002)	+0.707 (0.0002 → 0.0002)	+0.704 (0.0002 → 0.0002)
Δcross at recessions, GDP weights	4.17×10^{-5} (0.0002 → 0.0002)	4.48×10^{-5} (0.0002 → 0.0002)	5.72×10^{-5} (0.0002 → 0.0002)
Δown at recessions, GDP weights	**7.78×10^{-6} (0.038 → 0.053)**	8.96×10^{-6} (0.035 → 0.047)	1.41×10^{-5} (0.0002 → 0.0002)
Share of the excess from the cross term, GDP weights	+0.843 (0.316 → 0.319)	+0.833 (0.319 → 0.313)	+0.802 (0.344 → 0.349)
Cross share at recessions, equal weights	+0.783 (0.0002 → 0.0002)	+0.771 (0.0002 → 0.0002)	+0.876 (0.0002 → 0.0002)
Δcross at recessions, equal weights	4.72×10^{-5} (0.0002 → 0.0002)	4.97×10^{-5} (0.0002 → 0.0002)	5.22×10^{-5} (0.0002 → 0.0002)
Δown at recessions, equal weights	1.46×10^{-6} (0.0002 → 0.0004)	1.65×10^{-6} (0.0002 → 0.0004)	1.88×10^{-6} (0.0002 → 0.0002)
Share of the excess from the cross term, equal weights	+0.970 (0.478 → 0.476)	+0.968 (0.466 → 0.459)	+0.965 (0.476 → 0.466)
MST contraction before recessions	−0.157 (0.206 → 0.185)	−0.143 (0.219 → 0.216)	−0.171 (0.168 → 0.166)
MST contraction, detrended	+0.014 (0.561 → 0.524)	+0.057 (0.659 → 0.624)	−0.057 (0.356 → 0.338)
MST fragility before recessions	+0.129 (0.271 → 0.252)	+0.086 (0.338 → 0.336)	+0.100 (0.311 → 0.299)
Slowing down: autocorrelation of r	−0.152 (0.789 → 0.770)	−0.098 (0.676 → 0.678)	+0.027 (0.456 → 0.456)
Slowing down: variance of r	+0.438 (0.031 → 0.029)	+0.420 (0.037 → 0.037)	+0.438 (0.033 → 0.036)
Slower recovery after recessions	+0.002 (0.502 → 0.478)	+0.024 (0.507 → 0.488)	+0.078 (0.534 → 0.517)
Real-GDP channel arc, lead-1	+0.254 (0.019 → 0.020)	+0.245 (0.019 → 0.023)	+0.237 (0.019 → 0.025)
Real-GDP channel arc, crisis year	**+0.208 (0.094 → 0.046)**	**+0.202 (0.094 → 0.050)**	+0.192 (0.094 → 0.061)
Real-GDP channel arc, after-2	−0.320 (0.019 → 0.018)	−0.291 (0.019 → 0.029)	−0.332 (0.019 → 0.015)

Table A5. Multiplicity

Raw one-sided p, Bonferroni-adjusted p and Benjamini–Hochberg q over the five lags of the arc.

Lag	S31 raw p	S31 Bonferroni	S31 BH q	S30 raw p	S30 Bonferroni	S30 BH q	S26 raw p	S26 Bonferroni	S26 BH q
Lead-2	0.038	0.189	0.094	0.038	0.189	0.094	0.038	0.189	0.063
Lead-1	0.019	0.094	0.094	0.019	0.094	0.094	0.019	0.094	0.047
Crisis year	0.189	0.943	0.236	0.189	0.943	0.236	0.189	0.943	0.236
After-1	0.717	1.000	0.717	0.755	1.000	0.755	0.679	1.000	0.679
After-2	0.057	0.283	0.094	0.057	0.283	0.094	0.019	0.094	0.047

Note. With 53 circular shifts the smallest raw p is $1/53 = 0.019$, so the smallest Bonferroni-adjusted value over five lags is $5/53 = 0.094$: no single lag can pass Bonferroni at 0.05. Under Benjamini–Hochberg a lag at the floor passes only if a second lag is also at the floor (or four lags are at 0.038 or below), which happened on the original 26 (lead-1 and after-2) and not on the 30 or the 31. Corrected significance is therefore not what supports lead-1; its support is that it survives past-only phases and the removal of any one economy, and it rests on 2009 and 2020.

Table A6. Mean synchronization by period type and around recessions

Mean r with 95% bootstrap intervals (10,000 draws), and the superposed epoch: mean r from four years before to four years after the five recessions. Descriptive.

Sample	Recession years	Downturn years (1998, 2001, 2012)	Other years	Epoch mean r, t−4 ... t+4
S31	0.60 [0.39, 0.83]	0.30 [0.13, 0.55]	0.48 [0.42, 0.55]	0.49, 0.55, 0.64, 0.65, 0.60, 0.56, 0.31, 0.29, 0.35
S30	0.61 [0.39, 0.84]	0.30 [0.12, 0.55]	0.49 [0.42, 0.56]	0.50, 0.56, 0.64, 0.65, 0.61, 0.57, 0.33, 0.31, 0.37
S26	0.65 [0.45, 0.86]	0.37 [0.23, 0.62]	0.54 [0.47, 0.61]	0.57, 0.62, 0.70, 0.70, 0.65, 0.60, 0.35, 0.35, 0.40

Table A7. Effective number of independent economies

$N_{\text{eff}} = 1/E[r^2]$ over each group of years (Box 1), with two-sided (P) and past-only (C) phases; the year before means the years before the five recessions. Descriptive.

Sample, phases	All years	Recession years	Year before	Normal years	2009	2020
S31, two-sided (P)	3.46	2.36	2.06	3.50	1.37	1.11
S31, past-only (C)	5.97	2.18	2.59	7.17	1.38	1.15
S30, two-sided (P)	3.39	2.30	2.07	3.44	1.38	1.07
S30, past-only (C)	5.80	2.23	2.59	6.89	1.40	1.15
S26, two-sided (P)	2.88	2.08	1.86	2.90	1.25	1.04
S26, past-only (C)	4.78	1.75	2.10	5.80	1.17	1.00

By recession year, 31 economies:

Phases	1975	1982	1991	2009	2020
two-sided (P)	5.09	5.41	9.83	1.37	1.11
past-only (C)	2.72	6.48	5.71	1.38	1.15

Table A8. The variance and autocorrelation trends by recession

The early-warning trends of §3.5.7 with each recession alone, left out, or in pairs (31 economies; same surrogates and seed as the pre-registered test). Not pre-registered; asked for after the main run.

Recessions used	Variance trend τ (p)	Autocorrelation trend τ (p)
1982, 1991, 2009, 2020	+0.438 (0.031)	−0.152 (0.789)
1982	+0.393 (0.205)	+0.250 (0.317)
1991	−0.357 (0.668)	−0.643 (0.888)
2009	+0.929 (0.039)	−0.071 (0.555)
2020	+0.786 (0.117)	−0.143 (0.597)
1991, 2009, 2020	+0.452 (0.065)	−0.286 (0.874)
1982, 2009, 2020	+0.702 (0.001)	+0.012 (0.491)
1982, 1991, 2020	+0.274 (0.206)	−0.179 (0.763)
1982, 1991, 2009	+0.321 (0.125)	−0.155 (0.757)
1982, 1991	+0.018 (0.513)	−0.196 (0.748)
2009, 2020	+0.857 (0.003)	−0.107 (0.644)

Table A9. Rolling windows kept inside 1970–2022

Rolling statistic, windows kept inside 1970–2022	S31	S30	S26
Cross share of the recession excess, GDP / equal weights	86.8% / 96.9%	85.9% / 96.7%	82.4% / 96.9%
MST contraction before recessions, raw; detrended	+0.095 (0.654); +0.333 (0.910)	+0.119 (0.692); +0.381 (0.936)	+0.000 (0.504); +0.167 (0.740)
Autocorrelation trend	−0.107 (0.644)	−0.036 (0.550)	−0.071 (0.595)
Variance trend	+0.857 (0.003)	+0.857 (0.005)	+0.750 (0.019)

Rolling statistic, windows kept inside 1970–2022	S31	S30	S26
Recovery-rate change	+0.065 (0.520)	+0.102 (0.530)	+0.182 (0.578)

Table A10. Channels inside the economies

Synchronization of three channels from the Global Macro Database (an economy enters a channel only with continuous data from 1965), with two-sided phases and with phases from past data only. Description only: no channel lead survives past-only phases except real GDP's lead-1.

Channel	Economies	Lead-2	Lead-1	Crisis year	After-2	Past-only lead-2	Past-only lead-1
Real GDP (Global Macro Database)	30	+0.225 (0.057)	+0.254 (0.019)	+0.208 (0.094)	−0.320 (0.019)	−0.082 (0.679)	+0.319 (0.038)
Investment	28	+0.270 (0.038)	+0.262 (0.057)	+0.197 (0.132)	−0.203 (0.094)	−0.110 (0.717)	+0.178 (0.170)
Short-term interest rate	19	+0.273 (0.019)	+0.130 (0.113)	−0.021 (0.623)	−0.069 (0.226)	+0.165 (0.151)	−0.003 (0.472)

Table A11. G1b: the escape layer, every setting and sensitivity

Past-only lead-1 on 1,000 simulated worlds per row (§5.4). Differences in percentage points, paired by world.

Worlds	Layer	Past-only lead-1 rejects (95% interval)	Minus constant noise, points (95% interval)	Minus shuffled synchrony, points (95% interval)	Worlds at $p \leq 0.038$	Verdict
trade-coupled (κ*)	primary	6.3% (5.0%–8.0%)	+3.5 (+1.9 to +5.1)	+3.4 (+1.7 to +5.1)	6.3%	partly
trade-coupled (κ*)	one state per economy	3.8% (2.8%–5.2%)	+0.1 (−1.5 to +1.7)	+0.6 (−1.0 to +2.2)	3.8%	not reproduced
trade-coupled (κ*)	noise from Ā·R_A	7.1% (5.7%–8.9%)	+4.3 (+2.6 to +6.0)	+3.8 (+2.0 to +5.6)	7.1%	partly
trade-coupled (κ*)	μ = 0.1	6.7% (5.3%–8.4%)	+3.7 (+2.0 to +5.4)	+3.1 (+1.3 to +4.9)	6.7%	partly
trade-coupled (κ*)	μ = 0.3	5.8% (4.5%–7.4%)	+1.9 (+0.2 to +3.6)	+1.8 (+0.0 to +3.6)	5.8%	partly
trade-coupled (κ*)	no refractory period	8.8% (7.2%–10.7%)	+5.0 (+3.1 to +6.9)	+5.4 (+3.6 to +7.2)	8.8%	partly
trade-coupled (κ*)	refractory 4 years	4.2% (3.1%–5.6%)	+0.6 (−1.0 to +2.2)	+1.0 (−0.6 to +2.6)	4.2%	not reproduced
trade-coupled (κ*)	1 unit = 0.5 years	7.1% (5.7%–8.9%)	+4.4 (+2.6 to +6.2)	+3.3 (+1.4 to +5.2)	7.1%	partly
trade-coupled (κ*)	1 unit = 2 years	5.8% (4.5%–7.4%)	+1.7 (+0.0 to +3.4)	+2.3 (+0.7 to +3.9)	5.8%	partly
no coupling	primary	4.0% (3.0%–5.4%)	−0.5 (−2.1 to +1.1)	+0.0 (−1.6 to +1.6)	4.0%	not reproduced
common forcing only (φ*)	primary	3.2% (2.3%–4.5%)	−0.9 (−2.4 to +0.6)	−0.9 (−2.4 to +0.6)	3.2%	not reproduced

Table A12. The arc under every phase convention

Every lag, three samples (Table 2 gives the summary for the 31).

Sample	Convention	Lead-2	Lead-1	Crisis year	After-1	After-2	r–Ā detrended
S31	P	+0.235	+0.231	+0.159	+0.099	−0.258	+0.646
		(0.038)	(0.019)	(0.189)	(0.717)	(0.057)	(0.019)
S31	W	+0.216	+0.204	+0.134	+0.109	−0.165	+0.566
		(0.057)	(0.019)	(0.189)	(0.774)	(0.113)	(0.019)
S31	N	+0.216	+0.192	+0.135	+0.105	−0.190	+0.547
		(0.038)	(0.057)	(0.208)	(0.755)	(0.094)	(0.019)
S31	E	+0.233	+0.236	+0.159	+0.101	−0.160	+0.583
		(0.038)	(0.019)	(0.189)	(0.755)	(0.132)	(0.019)
S31	C	−0.044	+0.379	+0.417	+0.214	−0.159	+0.726
		(0.585)	(0.038)	(0.019)	(0.943)	(0.170)	(0.019)
S30	P	+0.226	+0.217	+0.161	+0.111	−0.236	+0.641
		(0.038)	(0.019)	(0.189)	(0.755)	(0.057)	(0.019)
S30	W	+0.204	+0.184	+0.130	+0.121	−0.131	+0.549
		(0.075)	(0.057)	(0.226)	(0.774)	(0.170)	(0.019)
S30	N	+0.204	+0.173	+0.133	+0.116	−0.158	+0.535
		(0.019)	(0.094)	(0.189)	(0.774)	(0.132)	(0.019)
S30	E	+0.224	+0.226	+0.164	+0.107	−0.136	+0.577
		(0.038)	(0.019)	(0.170)	(0.755)	(0.170)	(0.019)
S30	C	−0.021	+0.369	+0.397	+0.200	−0.161	+0.723
		(0.528)	(0.038)	(0.019)	(0.925)	(0.170)	(0.019)
S26	P	+0.230	+0.216	+0.149	+0.078	−0.277	+0.608
		(0.038)	(0.019)	(0.189)	(0.679)	(0.019)	(0.019)
S26	W	+0.210	+0.178	+0.112	+0.088	−0.144	+0.509
		(0.057)	(0.094)	(0.208)	(0.698)	(0.170)	(0.019)
S26	N	+0.210	+0.178	+0.124	+0.089	−0.197	+0.531
		(0.038)	(0.094)	(0.189)	(0.679)	(0.094)	(0.019)
S26	E	+0.228	+0.220	+0.152	+0.075	−0.170	+0.536
		(0.057)	(0.019)	(0.170)	(0.717)	(0.113)	(0.019)
S26	C	+0.007	+0.374	+0.446	+0.154	−0.167	+0.682
		(0.453)	(0.038)	(0.019)	(0.868)	(0.151)	(0.019)

Table A13. Test size in every G1 setting

The share of 1,000 simulated worlds in which each test rejects at $p \leq 0.05$, for each model (K: Kuramoto; SL: Stuart–Landau) and coupling (none, common forcing φ^*, trade links at half and at the full calibrated κ^*). **Bold**: the 95% interval lies above the correct rate (3.8% for circular shifts, 5% for 199 surrogates). Only the "none" columns measure size; the others show what each test does when coupling is there. Table 1 (§3.4.3) is the "none" columns with both dating rules.

Real recession dates.

Test	K none	K forcing	K κ*/2	K κ*	SL none	SL forcing	SL κ*/2	SL κ*
Arc, lead-2	3.3%	4.2%	3.6%	2.3%	3.6%	3.7%	4.7%	4.5%
Arc, lead-1	2.9%	3.8%	4.4%	3.6%	4.4%	**5.2%**	**5.3%**	4.0%
Arc, crisis year	4.4%	3.7%	3.1%	3.0%	3.3%	3.3%	3.0%	3.3%
Arc, after-1	4.5%	4.4%	4.3%	**5.9%**	3.6%	3.8%	4.5%	4.4%
Arc, after-2	**6.2%**	**7.0%**	**6.6%**	**11.6%**	4.6%	**6.4%**	**7.7%**	**6.4%**
Arc, lead-2, past-only phases	4.1%	4.1%	3.1%	3.1%	3.4%	3.2%	3.9%	3.4%
Arc, lead-1, past-only phases	3.5%	3.9%	4.2%	2.9%	3.8%	4.2%	4.6%	3.5%
r with swing size (shifts)	**6.2%**	**96.2%**	**25.1%**	**74.6%**	**5.7%**	**99.9%**	**66.5%**	**98.2%**
r with swing size (surrogates)	5.5%	**98.1%**	**26.1%**	**81.1%**	5.4%	**100.0%**	**74.5%**	**98.8%**
Cross term at recessions, GDP weights	5.3%	**99.9%**	**56.8%**	**97.1%**	5.4%	**100.0%**	**88.8%**	**99.3%**
Cross term at recessions, equal weights	4.6%	**100.0%**	**89.6%**	**100.0%**	4.6%	**100.0%**	**99.4%**	**100.0%**
Identity: alignment at recessions	4.8%	0.8%	3.7%	0.8%	**6.5%**	1.2%	2.7%	1.6%
Identity: swing size at recessions	**13.8%**	**27.1%**	**15.5%**	**21.0%**	**18.9%**	**29.4%**	**30.0%**	**35.6%**
MST contraction before recessions	2.8%	5.1%	**7.5%**	**17.3%**	4.1%	**6.7%**	**7.5%**	**7.2%**
MST contraction, detrended	3.0%	3.9%	4.9%	**6.5%**	4.5%	5.3%	5.9%	4.3%
MST fragility before recessions	4.9%	5.7%	**7.2%**	**15.3%**	5.2%	5.2%	**9.2%**	**8.2%**
Phase leads clustered	3.0%	**100.0%**	**64.6%**	**99.7%**	5.3%	**100.0%**	**95.6%**	**100.0%**
Slowing down: autocorrelation of r	5.1%	3.8%	5.8%	5.8%	4.2%	3.3%	6.0%	3.9%
Slowing down: variance of r	5.1%	5.1%	**9.4%**	**6.7%**	6.1%	4.9%	**7.4%**	**7.5%**
Slower recovery after recessions	4.8%	0.7%	0.9%	0.5%	4.5%	0.7%	1.3%	1.3%
Any of the five arc lags	18.4%	20.2%	18.7%	22.8%	17.3%	19.3%	20.2%	19.5%
Any of lead-2, lead-1, crisis	9.9%	10.9%	9.8%	7.8%	10.2%	10.8%	11.0%	10.7%

Each world's own recession dates.

Test	K none	K forcing	K κ*/2	K κ*	SL none	SL forcing	SL κ*/2	SL κ*
Arc, lead-2	**5.2%**	**7.4%**	3.0%	3.3%	**5.3%**	**8.9%**	3.6%	3.7%
Arc, lead-1	**8.5%**	**32.5%**	**10.8%**	**15.0%**	**8.6%**	**41.9%**	**16.3%**	**20.6%**
Arc, crisis year	**8.9%**	**29.6%**	**13.5%**	**16.8%**	**8.2%**	**38.1%**	**19.1%**	**25.1%**
Arc, after-1	4.2%	1.5%	2.6%	0.8%	2.6%	1.7%	2.0%	3.5%
Arc, after-2	3.9%	**5.7%**	**5.9%**	3.8%	4.3%	**6.2%**	**5.7%**	**5.9%**
Arc, lead-2, past-only phases	3.3%	2.4%	1.9%	1.3%	3.5%	3.3%	3.1%	1.4%
Arc, lead-1, past-only phases	2.9%	2.5%	1.8%	1.7%	2.3%	1.8%	2.0%	0.8%
r with swing size (shifts)	**6.2%**	**96.2%**	**25.1%**	**74.6%**	**5.7%**	**99.9%**	**66.5%**	**98.2%**
r with swing size (surrogates)	5.5%	**98.1%**	**26.1%**	**81.1%**	5.4%	**100.0%**	**74.5%**	**98.8%**
Cross term at recessions, GDP weights	**8.9%**	**100.0%**	**60.1%**	**98.1%**	**10.8%**	**100.0%**	**90.5%**	**99.1%**
Cross term at recessions, equal weights	**6.8%**	**100.0%**	**83.9%**	**100.0%**	**6.8%**	**100.0%**	**99.4%**	**100.0%**
Identity: alignment at recessions	**11.4%**	**11.2%**	**13.1%**	5.5%	**12.1%**	**17.9%**	**18.7%**	**11.7%**
Identity: swing size at recessions	**7.5%**	**62.0%**	**13.6%**	**43.8%**	**14.9%**	**75.1%**	**65.7%**	**86.9%**
MST contraction before recessions	5.0%	**9.3%**	**8.5%**	**28.5%**	3.8%	**10.1%**	**8.2%**	**12.6%**
MST contraction, detrended	5.1%	**6.5%**	5.8%	**10.2%**	3.7%	**6.8%**	5.2%	5.8%
MST fragility before recessions	5.3%	**8.2%**	**6.9%**	**23.9%**	5.1%	**9.8%**	**9.8%**	**11.8%**
Phase leads clustered	3.0%	**100.0%**	**64.6%**	**99.7%**	5.3%	**100.0%**	**95.6%**	**100.0%**
Slowing down: autocorrelation of r	5.7%	4.2%	5.0%	**6.6%**	4.5%	3.4%	5.0%	4.0%
Slowing down: variance of r	5.0%	5.6%	**10.1%**	5.8%	5.5%	5.9%	**7.9%**	**7.2%**
Slower recovery after recessions	5.2%	0.9%	1.4%	0.7%	4.0%	1.4%	1.5%	1.5%
Any of the five arc lags	26.2%	57.3%	29.8%	31.1%	25.0%	65.8%	37.6%	46.0%
Any of lead-2, lead-1, crisis	20.2%	54.0%	23.4%	27.5%	20.1%	62.9%	32.3%	39.8%

Supplementary figures

Redrawn for this version (31 economies, convention P): F1.3, F2.5, F4.1, F4.4, F5.1 and F6.1 (main text); F4.5 (both weightings) and F5.4 (Methods diagnostics). Every other figure in the project's figure catalogue (`docs/FIGURES.md`) predates the 31-economy sample and is supplementary only; its status there says whether it uses the 26 economies, convention W or superseded data.